

Lepton-Number-Violating Target Extension of the Neutrino Standing-Carrier Space

⁷⁶Ge search-response lift fibres, nuclear-interface descent,
and universal conjugation-coordinate splitting

PCR-5D2

Publication Final

Principal theorem

Let $\widehat{T}_{D2}^{\text{Ge}}$ extend the fixed oscillation target by the checksum-locked LEGEND-200, GERDA, and MAJORANA DEMONSTRATOR ⁷⁶Ge search response. For every PCR-5B/B15 framed physical realization W_H , the source-first compiled domain contains every $P_{H,C}$, $C \in \mathcal{V}_{76}^{\text{src}}$, with exact total amplitude $A_H + C$.

On the physical locus $A_H + C \in \mathcal{A}_{76}^{\text{phys}}$, the Constraint-Preserving Physical Amalgamation and Source-Faithful Enriched Realization Theorem constructs

$$W_{H,C} := W_H \bigcup_{\mathcal{B}_{76}(W_H)}^{\text{phys}} \mathcal{D}_{76}(H, C) \in \text{PhysMod}_{4,D2}^{\text{sem}},$$

with

$$\chi_{\text{car}}(W_{H,C}) \simeq P_{H,C}^{\text{red}}, \quad \Pi_{76}^{\text{tr}}(W_{H,C}) = A_H + C, \quad \text{Res}_{\text{osc}}(W_{H,C}) \simeq W_H.$$

Inherited, pure-process, cross, and common-cone residuals all vanish on this one semantic physical apex, while

$$\text{Del}_{c76}^{\text{phys}}(W_{H,C}) \simeq W_H, \quad \chi_{\text{car}} \circ \text{Del}_{c76}^{\text{phys}} \simeq U_{D2}^{\text{CMP}} \circ \chi_{\text{car}}.$$

The Retained-Oscillation Transparency and Base-Conservativity Theorem evaluates the fixed oscillation interface *with the correction physically present*, without invoking correction deletion, and proves

$$\boxed{\Phi_4^{\text{ret}}(W_{H,C}) = \Phi_4(W_H)}, \quad \boxed{\Xi_{\text{osc}}^{\text{ret}}(W_{H,C}) = \Xi_{\text{osc}}(W_H)}$$

for every physically admissible compiled correction. Thus the compiled physical locus is genuinely framed over the same oscillation standing core, not merely retractable to it after the new process is removed.

Once the source, physical, and retained-base structures are fixed, the response analysis selects (A_{76}^*, η_*) . The marked correction $C_{H,z}^* = -A_H + zA_{76}^*$ lies in the transparent locus and gives an active continuum interval of base-preserving survivors and therefore

$$\boxed{\text{im } \rho_{D2} = Q_{\text{osc}}^{\text{SCIC}}}, \quad \boxed{|\pi_0 \mathcal{L}_{D2}(q)| \geq \mathfrak{c} \quad \forall q \in Q_{\text{osc}}^{\text{SCIC}}}.$$

No base class is eliminated; every base class splits into at least continuum many D2-standing-distinct enriched classes. The new coordinate is conjugation/LNV response, not forced Majorana character.

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Abstract

PCR-5B identifies the standing-carrier quotient of the fixed neutrino-oscillation target as an exact mixed moduli object with at least continuum many SCIC classes. PCR-5C supplies the target-extension calculus, and PCR-5D1 shows that direct kinematic mass preserves the full oscillation base while introducing continuum absolute-scale multiplicity. The present analysis applies that calculus to the same-isotope ^{76}Ge neutrinoless-double-beta-decay response determined by LEGEND-200, GERDA, and the MAJORANA DEMONSTRATOR.

The empirical primitive is the signal-rate or half-life response, together with the detector model, likelihood, and nuisance structure; it is not the model label “Majorana neutrino.” Non-observation is distinguished from physical zero. The familiar quantity $m_{\beta\beta} = |\sum_i U_{ei}^2 m_i|$ is used only on the light-Majorana-exchange locus, while the high-scale dimension-nine SMEFT response forms a separate branch. Nuclear matrix elements, detector calibration, background, and statistical nuisance variables remain response coordinates unless a standing theorem shows otherwise.

Every finite PCR-3/B15 source hypergraph admits a finite exact transition-amplitude normal form $A_H = \mathcal{A}_{76}^{\text{src}}(H) \in \mathcal{V}_{76}^{\text{src}}$. The source-defined correction space $\mathcal{V}_{76}^{\text{src}}$ is fixed independently of target evaluation, and the physical amplitudes form the contraction locus $\mathcal{A}_{76}^{\text{phys}}$. Each $A \in \mathcal{A}_{76}^{\text{phys}}$ has an exact Julia–Halmos scattering realization whose observed transition block is precisely A .

For a framed PCR-5B/B15 physical realization W_H and a correction C with $A_H + C \in \mathcal{A}_{76}^{\text{phys}}$, the source-faithful enriched realization is the split-anchored physical amalgamation

$$W_{H,C} = W_H \bigcup_{\mathcal{B}_{76}(W_H)}^{\text{phys}} \mathcal{D}_{76}(H, C).$$

Its activity-reduced carrier trace is $P_{H,C}^{\text{red}}$, its exact D2 transition block is $A_H + C$, and its physical restriction is W_H . The complete PCR-4/D2 residual family separates into inherited, pure-process, cross, and common-cone sectors, each of which vanishes on the amalgamated object. Physical correction deletion returns W_H without retuning, and the source reduct is the carrier-trace image of that physical restriction.

The retained-oscillation theorem evaluates the complete PCR-4/SCIC oscillation trace directly on $W_{H,C}$ while the D2 correction remains present. The D2-only cells generate a target-relative ideal annihilated by every fixed oscillation evaluator and licensed continuation. Consequently,

$$\Phi_4^{\text{ret}}(W_{H,C}) = \Phi_4(W_H), \quad \Xi_{\text{osc}}^{\text{ret}}(W_{H,C}) = \Xi_{\text{osc}}(W_H)$$

channel by channel, without using correction deletion or base retraction.

After the source, physical, and retained-base structures are fixed, a response-visible pair (A_{76}^*, η_*) determines the family $C_{H,z}^* = -A_H + zA_{76}^*$. The corresponding source-faithful objects have exact transition block zA_{76}^* , framewise kernel $zK_{*,\eta}$, and rate $z^2\gamma_{*,\eta}$. A nondegenerate accepted interval yields continuum many standing-distinct lifts over every base class. The resulting endpoint is

$$\text{im } \rho_{D2} = Q_{\text{osc}}^{\text{SCIC}}, \quad |\pi_0 \mathcal{L}_{D2}(q)| \geq \mathfrak{c} \quad \forall q \in Q_{\text{osc}}^{\text{SCIC}}.$$

Every fibre has coarse type `ParameterizedMixed`; no comparison-class-independent upper cardinal is claimed. The present null search neither forces a nonzero physical LNV response nor selects Majorana character.

Scope and interpretation

The result is relative to the fixed ^{76}Ge data release, the PCR-3 source comparison class, PCR-4 semantic one-point survival, PCR-5B/B15, PCR-5C, the explicitly declared exact-amplitude extension SOE-D2-1, and the accompanying target/source specification. The exact-amplitude colligation is a target-scale process realization, not a unique local ultraviolet completion, and it does not replace the retained oscillation carrier. Intrinsic local-QFT charts retain all black-box-induced self-energy, matching, and continuation consequences before the correction amplitude is added. The conclusions depend on amplitude-normal-form coverage, contraction-space physicality, exact-amplitude realization, constraint-preserving split attachment, activity reduction, in-situ retained-

oscillation transparency, semantic common-cone realization, deletion naturality, response descent, standing-amplitude descent, family classification, and overlap closure. A counterexample to any of these stated hypotheses would revise the corresponding conclusion. Supplementary hashes and machine-readable records establish provenance and reproducibility; the mathematical and physical claims rest on the arguments given here.

Nonclaim 0.1 (No amplitude-only replacement of the enriched carrier). *The proof does not identify an interaction-Hamiltonian derivative with a finite transition amplitude, does not multiply separately realized scattering processes, and does not replace the retained PCR-5B physical carrier by an aggregate D2 colligation. The complete amplitude $A_H + C$ is the D2 transition leg of one source-faithful physical object $W_{H,C}$ whose base restriction is the original realization W_H .*

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1 Introduction and principal results

1.1 The D2 target-extension problem

The base object is the exact oscillation standing-carrier quotient $Q_{\text{osc}}^{\text{SCIC}}$, rather than a list of preferred microscopic models. A lawful target extension acts through a restriction functor

$$R_{D2} : \mathcal{N}(\hat{T}_{D2}^{\text{Ge}}) \longrightarrow \mathcal{N}(T_{\text{osc}}), \quad \rho_{D2} = \pi_0(R_{D2}) : Q(\hat{T}_{D2}^{\text{Ge}}) \longrightarrow Q_{\text{osc}}^{\text{SCIC}},$$

and through the connected-subgroupoid lift groupoid

$$\mathcal{L}_{D2}(q) = \mathcal{N}(\hat{T}_{D2}^{\text{Ge}}) \times_{\mathcal{N}(T_{\text{osc}})}^h \mathcal{N}(T_{\text{osc}})[q].$$

For each $q \in Q_{\text{osc}}^{\text{SCIC}}$, the D2 calculation must determine whether the lift fibre is empty, singleton, finitely or countably split, or a continuous, functional, categorical, or mixed moduli object. The image of ρ_{D2} , the new standing separator, and the S/R/P/Y source-composition profile are logically separate outputs.

The phenomenon target is the present ^{76}Ge neutrinoless-double-beta-decay search response: the nuclear transition, event kernel, apparatus response, released likelihood, nuisance and covariance structure, and exact non-observation semantics. Light-Majorana exchange and high-scale EFT are specialized response branches rather than universal definitions of the target.

1.2 Null-search semantics

A null search constrains rates; it does not instantiate a physical zero. A D2 lift survives when its complete defined response lies in the accepted target region, including a defined zero-rate locus. Undefined response, response out of scope, certified N/A, source update, and target update are distinct statuses and cannot be reclassified as accepted zero-rate points. In particular,

$$\text{NoEvidence} \not\Rightarrow \text{PhysicalZero}, \quad \text{Undefined} \not\Rightarrow \text{AcceptedZero}.$$

Universal base survival therefore cannot be inferred by declaring that every base class has “no LNV.” It requires a source-generated, physically typed enriched realization over every base frame.

1.3 Source-faithful enriched realization

An exact aggregate transition amplitude does not by itself specify an enriched carrier. The analysis therefore distinguishes

$$\begin{aligned} P_{H,C} & \quad \text{the compiled source,} \\ \text{RealAmp}_{76}(A_H + C) & \quad \text{the exact D2 process leg,} \\ W_{H,C} & \quad \text{the source-faithful enriched realization.} \end{aligned}$$

For a framed semantic base realization W_H , the D2 cells attach by closed split isometries along the retained boundary and owner diagram $\mathbf{B}_{76}(W_H)$. The relative physical amalgamation

$$W_{H,C} = W_H \bigcup_{\mathbf{B}_{76}(W_H)}^{\text{phys}} \mathbf{D}_{76}(H, C)$$

contains the base realization exactly once, together with the correction, the exact amplitude-sum relation, its unitary colligation, ownership, matching, continuation, induced effects, and every primitive or higher D2

incidence. Matching, continuation, vacuum and deformation, causal, induced-effect, and higher-incidence relations are evaluated on this common object, which lies in the semantic physical-model locus.

A target extension must also preserve the old standing structure while the new process is present. The fixed oscillation evaluator algebra is therefore applied directly to the enriched realization. The D2-only cells carry a relative target grade and generate an ideal with no projection to a fixed oscillation-output sort. The retained-oscillation theorem consequently shows that the old trace and all seven SCIC channels factor through the retained base block in situ.

Only after the complete source domain, physical realization, and retained-base theorem have been established is the response-visible amplitude A_{76}^* selected. The family $C_{H,z}^* = -A_H + zA_{76}^*$ is then a distinguished subfamily of the already complete source domain, and the corresponding survivor is the full transparent object $W_{H,z}^*$ rather than the amplitude colligation alone.

1.4 Structure of the argument

The proof follows the D2-01–D2-12 sequence:

1. D2-01 specifies the empirical, response, and status target.
2. D2-02 gives the intrinsic, compiled, and hybrid source universe with exact base reduct.
3. D2-03 proves source-faithful one-point survival and the accepted continuum family.
4. D2-04 constructs standing restriction and establishes no merger.
5. D2-05 classifies the lift fibre over every base class.
6. D2-06 descends nuclear, detector, statistical, chart, and presentation coordinates.
7. D2-07 determines the identity-preserving and target-survival deletion strata and the S/R/P/Y code.
8. D2-08 computes the exact base image.
9. D2-09 proves universal splitting and identifies the newly standing-visible coordinate.
10. D2-10 resolves pairwise and higher overlaps before codescent.
11. D2-11 classifies N/A, failure, undefinedness, response-out-of-scope, source-update, and target-update frontiers.
12. D2-12 supplies the marginal datum required for PCR-5E.

1.5 Principal result

For every B15 frame H , the source-amplitude normal form gives $A_H = \mathcal{A}_{76}^{\text{src}}(H) \in \mathcal{V}_{76}^{\text{src}}$. The compiled source domain is

$$\mathcal{C}_{\text{CMP}}^{\text{raw}}(H) = \{P_{H,C} \mid C \in \mathcal{V}_{76}^{\text{src}}\},$$

with total amplitude and framewise response

$$A_{H,C}^{\text{tot}} = A_H + C, \quad K_{H,C,\eta} = \mathcal{R}_{76,\eta}(A_H + C).$$

For each physical total amplitude, the constraint-preserving amalgamation gives $W_{H,C} \in \text{PhysMod}_{4,D2}^{\text{sem}}$ with activity-reduced carrier trace $P_{H,C}^{\text{red}}$, D2 transition block $A_H + C$, and physical base restriction W_H . In addition,

$$\Phi_4^{\text{ret}}(W_{H,C}) = \Phi_4(W_H), \quad \Xi_{\text{osc}}^{\text{ret}}(W_{H,C}) = \Xi_{\text{osc}}(W_H),$$

with the correction still present. The family $C_{H,z}^* = -A_H + zA_{76}^*$ then yields standing-distinct accepted points over every base core. Hence

$$\boxed{\text{im } \rho_{D2} = Q_{\text{osc}}^{\text{SCIC}}}, \quad \boxed{|\pi_0 \mathcal{L}_{D2}(q)| \geq \mathfrak{c} \quad \forall q \in Q_{\text{osc}}^{\text{SCIC}}}.$$

The six intrinsic families and the complete correction-amplitude chart are classified with their response, status, descent, deletion, and overlap data.

1.6 Interpretation and limits

D2 realizes the independent-coordinate branch of the post-D1 refinement: it preserves the full oscillation base and introduces a conjugation/LNV response coordinate within every enriched fibre. It does not establish a global product decomposition, a common generator with oscillation continuation, or a nonzero realized LNV amplitude. Nuclear-matrix-element schemes, operator charts, field counts, naturalness criteria, and provenance labels do not acquire standing authority. Intrinsic local-QFT mechanisms retain their induced consequences and are not replaced by the compiled process chart.

2 Dependencies and the no-backflow boundary

2.1 PCR-1 through PCR-4

PCR-1 fixes the oscillation role without assuming a mass mechanism, PMNS ontology, Dirac/Majorana character, sterile sector, or candidate particle. PCR-2 fixes the complete carrier-neutral source–propagation–detection interface and its shared-witness discipline. PCR-3 supplies source-first semantic candidate completeness for the fixed source ontology and an explicit reopening rule for genuinely new source sorts, operations, incidences, or constitutive equations. PCR-4 evaluates all active propagation, gauge, anomaly, vacuum, deformation, matching, generalized-structure, and common-cone relations on one physical point [1–4].

2.2 PCR-5 through PCR-5B

PCR-5 fixes one terminal complete-carrier role and separates internal factorization from terminal carrier identity. PCR-5A proves that the seven-channel packet is SCIC-complete and preserves four marked deletion classes. Standing Carrier Identity Closure fixes equality of complete continuation signatures as the unique same-scope carrier identity relation. PCR-5B then executes the exact LAW/LOCAL/HID/-COMP/SOL/HYB atlas, lawful groupoids, failure frontiers, and B15 provenance/overlap handoff [5–8].

2.3 PCR-5C target-extension calculus

PCR-5C supplies the category of target-complete objects, base-source conservativity, standing restriction, no merger, connected-subgroupoid homotopy fibres, lift pseudofunctor, Grothendieck reconstruction, composition, joint-target discipline, and S/R/P/Y source-composition calculus [9]. Its global decomposition is

$$Q(\hat{T}_{D2}^{\text{Ge}}) \cong \bigsqcup_{q \in Q_{\text{osc}}^{\text{SCIC}}} \pi_0 \mathcal{L}_{D2}(q),$$

once every D2 fibre is semantically exhausted.

2.4 PCR-5D1 and the refinement addendum

D1 establishes

$$\text{im } \rho_{\beta, BB} = Q_{\text{osc}}^{\text{SCIC}}, \quad |\pi_0 \mathcal{L}_{\beta, BB}(q)| \geq \mathfrak{c} \quad \forall q,$$

so absolute kinematic scale does not select the oscillation base. The refinement addendum therefore requires D2 to report not only its base image and fibre type but also the class of newly standing-visible coordinate and whether it is independent of, coupled to, or a common generator with oscillation continuation [10, 11].

2.5 Target and source specification

The five collaboration-released LEGEND CSV files and the target semantics are fixed independently of the carrier calculation. The accompanying target/source specification retains the empirical object and states the constraint-preserving split attachment, activity-reduced trace, and conservative exact-amplitude source extension. The source-defined correction space $\mathcal{V}_{76}^{\text{src}}$, physical contraction locus $\mathcal{A}_{76}^{\text{phys}}$, exact process leg RealAmp_{76} , source-faithful physical object $W_{H,C}$, physical restriction, deletion, and downstream response-frame groupoid are typed as distinct objects [12].

2.6 No-backflow theorem

Theorem 2.1 (Source-formation independence and response-mutation invariance). *For every framed base source H ,*

$$\mathcal{C}_{\text{CMP}}^{\text{raw}}(H) = \{P_{H,C} : C \in \mathcal{V}_{76}^{\text{src}}\}$$

is formed solely from source sorts, exact source-amplitude equations, the typed correction space, ownership, deletion, and the exact reduct. It is independent of every response map, response frame, Hermitian rate form, likelihood value, acceptance value, standing class, desired base image, and desired fibre cardinality. Deleting, replacing, or randomizing all downstream response objects leaves this raw source groupoid unchanged; only its later physicality and response/status partition may change.

Proof. PCR-3 supplies source independence for the intrinsic universe. The extension adds the response-blind parameter space $\mathcal{V}_{76}^{\text{src}}$ and the source presentation map $(H, C) \mapsto P_{H,C}$. The contraction test $A_H + C \in \mathcal{A}_{76}^{\text{phys}}$, the exact-amplitude realization, and the response evaluation occur only after raw source formation. Hence no response mutation changes candidate membership. The marked affine family is selected later as a proof witness from this fixed domain and has no source-formation authority. $\square$

The extension is a comparison-class declaration, not an AASC existence inference. AASC evaluates independently generated candidates; it does not create a correction amplitude, turn an empty candidate fragment into a process, or convert a target-facing kernel into source ontology.

3 The checksum-locked ^{76}Ge empirical target

3.1 Target tuple

The empirical target is

$$T_{76}^{\text{emp}} = (D_{\text{evt}}, D_{\text{part}}, Q_{\beta\beta}, \mathcal{I}_{\text{Ge}}, \mathcal{R}_{\text{det}}, \mathcal{B}, \mathcal{L}_{\text{prof}}, \mathcal{P}, \mathcal{A}_{90}, \mathcal{U}_{\text{emp}}),$$

where the entries respectively denote the released event list, partition table, transition energy, isotope/decay identity, detector response, background model, profile likelihood, released posterior objects, published acceptance relation, and empirical update boundary.

The transition is

$$^{76}\text{Ge} \longrightarrow ^{76}\text{Se} + 2e^-, \quad Q_{\beta\beta} = 2039.061(7) \text{ keV}.$$

The first LEGEND-200 search uses 61.0 kg yr; the combined same-isotope analysis incorporates 127.2 kg yr from GERDA and 64.5 kg yr from the MAJORANA DEMONSTRATOR, finds no evidence for a signal, and reports

$$T_{1/2}^{0\nu} > 1.9 \times 10^{26} \text{ yr} \quad (90\% \text{ C.L.})$$

[13–15].

3.2 Exact public-byte identity

The five collaboration files are retained without reserialization. Their official MD5 values are:

File	Bytes	Official MD5
ge76-comb-0vbb-rate-posterior.csv	4,958	57be2395db8820a178ad0e04a0ab77e7
l200-0vbb-rate-posterior.csv	4,956	ff577eaa46778fe4bdebb77ac6bad9ea
l200-event-list.csv	580	51261b2ed1d69777b61196e6c243c399
l200-partition-parameters.csv	157,063	054a92ce3542e839a6874bdc74366872
profile-likelihood.csv	45,907	4f5eba2369aa5e60ef0de3a83daaab40

The supplementary integrity record gives the corresponding SHA-256 values. Byte identity establishes provenance, whereas target completeness follows from the target specification and response analysis.

3.3 Consistency checks on the released record

The partition file contains 316 rows and 72 detector labels. Summing the released exposure column gives 60.9810431 kg yr, consistent with the rounded 61.0 kg yr reported by the collaboration. The released event list contains 13 rows; applying the primary-paper exclusion of ± 5 keV around the 2119 keV gamma line leaves 11 rows. A discrepancy between one prose energy and the checksum-verified byte stream is recorded without alteration.

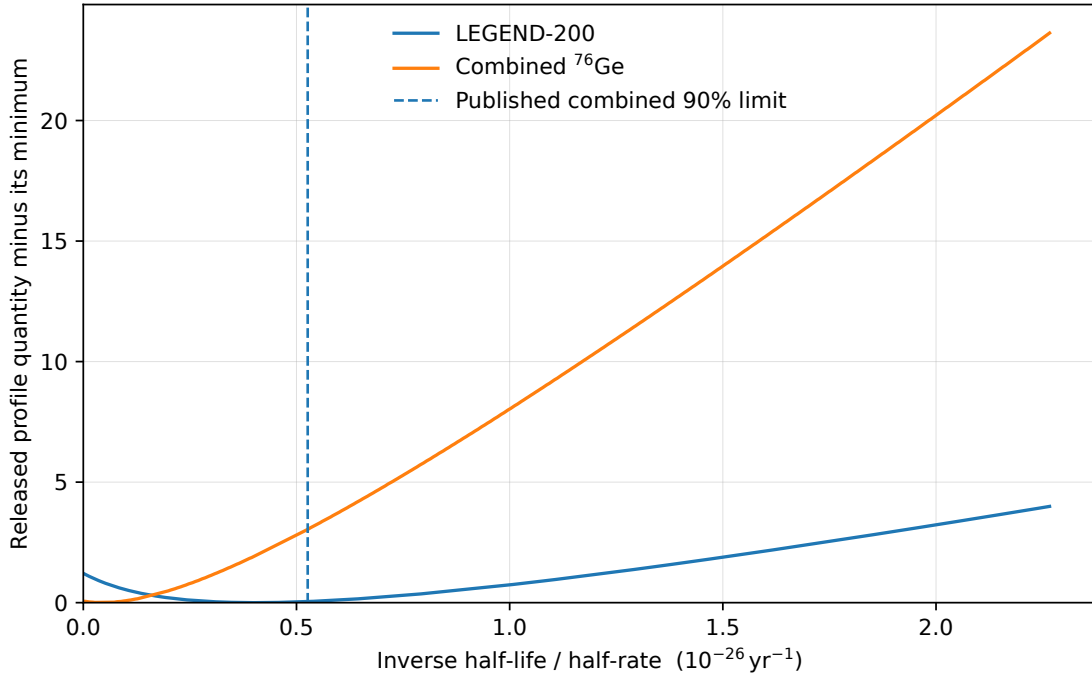


Figure 1: Released LEGEND-200 and combined ^{76}Ge profile-likelihood quantities after subtracting each released minimum. The vertical line marks the published combined 90% inverse-half-life bound Γ_{90}^{Ge} . The figure is a direct visualization of the public CSV and is not a reproduction of the collaboration's Monte Carlo confidence construction.

3.4 Carrier-neutral empirical coordinate

Define the nonnegative inverse half-life or half-rate

$$\Gamma_{1/2}^{0\nu} = (T_{1/2}^{0\nu})^{-1} \in \mathbb{R}_{\geq 0}.$$

The accepted interval adopted in the target specification is

$$J_{90}^{\text{Ge}} = [0, \Gamma_{90}^{\text{Ge}}), \quad \Gamma_{90}^{\text{Ge}} = (1.9 \times 10^{26} \text{ yr})^{-1}.$$

This interval is an empirical acceptance statement under the specified confidence construction. It does not identify the profile-likelihood value with a probability of a carrier class and does not make the upper endpoint a fundamental physical constant.

3.5 No-evidence and physical-zero separation

Definition 3.1 (D2 empirical statuses). *The target distinguishes:*

- *NoEvidence*: the status of the amplitude data analysis;
- *PhysicalZero*: an exact source-physical zero of the D2 kernel/rate;
- *TargetCompatible*: a defined response in the accepted locus;
- *AcceptanceFail*: a defined response outside the frozen acceptance locus;
- *Undefined*: an applicable response relation whose required object is unresolved;

- **ResponseOutOfScope**: *a meaningful source proposal not described by the frozen response target*;
- **CertifiedNA**, **SourceUpdate**, and **TargetUpdate**.

Proposition 3.2 (No null-search promotion). *There is no target-authorized implication*

$$\text{NoEvidence} \implies \text{PhysicalZero},$$

and hence no implication from the frozen empirical record alone to exact lepton-number conservation, Dirac character, or absence of a Majorana mass term.

Proof. The empirical statement is a finite-data likelihood result with a nonempty accepted interval containing positive rates. Physical zero is a source-theoretic equality. Since positive rates in J_{90}^{Ge} are target compatible, the empirical status does not entail the source equality. Promoting the former into the latter would add an undeclared classifier and would contradict the explicitly frozen status grammar. $\square$

3.6 Target update boundary

A new isotope, new LEGEND release, changed analysis window, different statistical construction, new detector response primitive, multi-isotope likelihood, or new final-state topology is a target update. Such a target may be evaluated later but does not retroactively change the theorem proved here.

4 Mechanism-neutral response and specialized branches

4.1 Complete response frames and fibred kernel spaces

A complete D2 response frame is

$$\eta = (\eta_{\text{nuc}}, \eta_{\text{det}}, \eta_{\text{stat}}) \in \mathcal{E}_{76}^{\text{adm}}.$$

The nuclear coordinate fixes one lawful exact-amplitude-to-nuclear-transition map

$$\mathcal{N}_{76}(\eta_{\text{nuc}}) : \mathcal{V}_{76}^{\text{src}} \longrightarrow \mathcal{A}_{76,\eta}^{\text{nuc}},$$

the detector coordinate fixes

$$\mathcal{D}_{76}(\eta_{\text{det}}) : \mathcal{A}_{76,\eta}^{\text{nuc}} \longrightarrow \mathbb{K}_{76,\eta},$$

and the statistical coordinate fixes the likelihood, profiling/marginalization prescription, covariance object, and acceptance rule used on that same physical response. Define

$$\boxed{\mathcal{R}_{76,\eta} := \mathcal{D}_{76}(\eta_{\text{det}}) \circ \mathcal{N}_{76}(\eta_{\text{nuc}})}. \quad (1)$$

The kernel fibre $\mathbb{K}_{76,\eta}$ retains coherent amplitude addition before detector trace and uses orthogonal summands for physically incoherent sectors.

Each complete frame carries a positive Hermitian response form

$$\mathfrak{S}_{76,\eta} : \mathbb{K}_{76,\eta} \times \mathbb{K}_{76,\eta} \longrightarrow \mathbb{C}, \quad \mathfrak{S}_{76,\eta}(K, K) \geq 0, \quad (2)$$

whose diagonal is the physical signal rate. It also carries a detector instrument $\mathcal{I}_{76,\eta}$, likelihood $\mathcal{L}_{76,\eta}$, and a total predicate $\text{Acc}_{90,\eta}(K) \in \{0, 1\}$ that includes response definedness and the fixed 90% acceptance semantics.

4.2 Response groupoid, covariance, and descent

The response groupoid $\Gamma_{76}^{\text{resp}}$ is generated by lawful nuclear-response chart changes, detector/calibration coordinate transformations, and statistical reparameterizations that preserve one complete physical response. An arrow $g : \eta \rightarrow \eta'$ supplies a kernel-space isomorphism

$$U_g : \mathbb{K}_{76,\eta} \xrightarrow{\sim} \mathbb{K}_{76,\eta'}$$

satisfying

$$U_g \mathcal{R}_{76,\eta} = \mathcal{R}_{76,\eta'}, \quad (3)$$

$$\mathfrak{G}_{76,\eta'}(U_g K_1, U_g K_2) = \mathfrak{G}_{76,\eta}(K_1, K_2), \quad (4)$$

$$\mathcal{J}_{76,\eta'}(U_g K) \simeq \mathcal{J}_{76,\eta}(K), \quad \text{Acc}_{90,\eta'}(U_g K) = \text{Acc}_{90,\eta}(K). \quad (5)$$

Theorem 4.1 (Fibred response covariance and descent). *Equations (3)–(5) define a lawful functor from the complete response-frame groupoid to physical response objects. The physical rate, likelihood ordering, definedness, and acceptance status descend to $|\mathcal{E}_{76}^{\text{adm}}/\Gamma_{76}^{\text{resp}}|$. A transformation that changes any descended physical output is not a nuisance arrow: it is a standing separator, source change, or target update.*

Proof. The generators of $\Gamma_{76}^{\text{resp}}$ are fixed precisely by covariance of the nuclear map, detector instrument, and statistical record. Their defining intertwiners preserve composition, identities, and inverses. Hermitian covariance gives equality of rates; the frozen likelihood and covariance transport give equality of likelihood ordering and acceptance. Therefore every displayed quantity is constant on groupoid orbits. A noncovariant change cannot be quotiented without identifying different target records, so it lies outside the nuisance groupoid. $\square$

4.3 Exact target-survival quantifier

The target does not compare separately chosen nuclear, detector, and statistical coordinates. One verdict uses one complete frame. Profiling or marginalization may occur only inside η_{stat} , with the correlated nuclear/detector maps held as members of that same frame. Define

$$\boxed{\text{Surv}_{76}^{\text{resp}}(A) = 1 \iff \exists[\eta] \in |\mathcal{E}_{76}^{\text{adm}}/\Gamma_{76}^{\text{resp}}| : \mathcal{R}_{76,\eta}(A) \text{ is defined and } \text{Acc}_{90,\eta}(\mathcal{R}_{76,\eta}(A)) = 1.} \quad (6)$$

This existential complete-frame semantics is part of the target freeze, not selected after the candidate or endpoint is known. It permits alternative lawful response realizations where the target genuinely leaves them open, but forbids taking a nuclear matrix element from one physical frame, a detector model from another, and a statistical verdict from a third.

4.4 EXP_RATE branch

The primary branch applies whenever a source point supplies a physical ^{76}Ge signal kernel or spectrum in at least one lawful complete response frame. For (X, η) ,

$$\text{Resp}_{\text{exp},\eta}(X) = ([K_{76,\eta}(X)]_{\text{resp}}, \mathfrak{G}_{76,\eta}(K_{76,\eta}(X), K_{76,\eta}(X)), \mathcal{J}_{76,\eta}(K_{76,\eta}(X)), \mathcal{L}_{76,\eta}, \text{Acc}_{90,\eta}(K_{76,\eta}(X))).$$

No microscopic Majorana assumption is required.

4.5 Light-Majorana-exchange branch

On the certified light-neutrino-exchange locus, the source supplies a self-conjugate light-neutrino propagator and

$$m_{\beta\beta} = \left| \sum_i U_{ei}^2 m_i \right|.$$

For each complete frame, the conventional comparator is

$$[T_{1/2}^{0\nu}]^{-1} = g_A^4 G_{0\nu}(\eta_{\text{nuc}}) |M_{0\nu}^{\text{eff}}(\eta_{\text{nuc}})|^2 \frac{|m_{\beta\beta}|^2}{m_e^2}.$$

The effective response includes the branch content required at the declared EFT order; a short-range contact contribution is already required at leading order for a renormalized light-exchange amplitude [16–18]. No one NME number is a universal carrier definition.

4.6 High-scale LNV EFT branch

For source points satisfying the high-scale SMEFT hypotheses, the target uses the complete amplitude vector and positive rate form of Cirigliano et al. [19]. Wilson coefficients, QCD evolution, hadronic low-energy constants, phase-space factors, nuclear matrix elements, and interference data belong to one branch response frame. The formula is not applied outside its hypotheses.

4.7 Nuclear-response object and applicability compiler

The nuclear coordinate contains

$$\mathcal{E}_{\text{nuc}} = (G_{0i}, M_i, g_i^{\pi\pi}, g_i^{\pi N}, g_i^{NN}, g_\nu^{NN}, \mathcal{U}_{\text{nuc}}, \Gamma_{\text{nuc}}).$$

The ab initio ^{76}Ge calculation of Belley et al. [20] supplies one uncertainty structure, not a carrier selector. For each branch r , applicability $A_r(X)$ is fixed before numerical evaluation, and

$$\text{Stat}_r(X) \in \{\text{NA}, \text{Pass}, \text{Fail}, \text{Undef}, \text{ROOS}, \text{SU}, \text{TU}\}.$$

An active failure cannot be laundered into N/A. A source outside a specialized chart is routed to EXP_RATE where possible; otherwise it receives response-out-of-scope or update status rather than physical exclusion.

Theorem 4.2 (D2-01: exact enriched target completeness). *The checksum-locked empirical record, complete response-frame groupoid, fibred maps $\mathcal{R}_{76,\eta}$, positive Hermitian forms $\mathfrak{G}_{76,\eta}$, instruments, likelihood/acceptance objects, EXP_RATE branch, LME and SMEFT_D9 specializations, applicability compiler, total status semantics, survival quantifier (6), and update boundary form a complete target object for the declared ^{76}Ge comparison scope.*

Proof. Every field used by the released likelihood belongs to a complete frame. Every in-scope proposal either produces a physical event kernel in a lawful frame or receives a typed undefined, response-out-of-scope, or update disposition. LME and SMEFT_D9 have explicit hypotheses. The response groupoid quotients only transformations preserving all descended physical outputs. The ordered applicability/status compiler gives exactly one disposition, while (6) fixes the candidate-level acceptance quantifier without cross-frame splicing. No untyped residual case remains inside the declared target. $\square$

D2-01 proves target completeness at the declared response scope. It does not prove that every raw source point has a defined response, passes the target, or belongs to a specialized branch.

5 Candidate-complete D2 source lifts

5.1 B15-framed intrinsic lifts

For each exact B15 source chart $a = (F, \sigma)$, let $H = H_a(\theta)$ be a framed source presentation over a concrete base standing core $n \in \mathcal{N}(T_{\text{osc}})$. An intrinsic D2 lift is a finite source-coherent hypergraph

$$X_{\text{int}} = (H; V_{D2}, E_{D2}, \omega_{D2}, \Theta_{D2}, a_{76})$$

formed by the fixed PCR-3 theory. Its source activity is fixed pointwise before target evaluation. The intrinsic universe contains effective laws, local and hidden carriers, composites, solitons, boundary/environmental sectors, and arbitrary finite hybrids. Nuclear response, detector response, likelihood, and calibration remain target objects unless a proposal independently makes them occupancy-bearing.

5.2 Finite exact D2 source-amplitude normal form

Let G_a be the finite activity-reduced source hypergraph of a framed B15 point. The fixed D2 boundary determines source input/output modules $\mathcal{H}_i^{76}$, $\mathcal{H}_f^{76}$, and a direct sum $\mathcal{E}_{76}$ of source-authorized unresolved labels. The amplitude coordinate is the dimensionless normalized transition block between these fixed asymptotic boundary modules. Dimensionful phase-space, nuclear, detector, exposure, and unit-conversion data enter only through the downstream maps $\mathcal{R}_{76,\eta}$ and forms $\mathfrak{G}_{76,\eta}$. The later norm bound $\|A\| \leq 1$ is therefore the exact probability-conservation condition for a unitary scattering block, not a naturalness, small-coupling, or candidate-ranking cut. For each source-active decorated hyperpath π , let $\text{Contr}_\pi(\theta)$ be the exact target-facing transition contraction obtained from the actual source cells. Define

$$A_{76,a}^{\text{src}}(\theta) = \bigoplus_{\lambda \in \Lambda_a^{\text{inc}}} \sum_{\pi \in \Pi_{76}^\lambda(a)} \text{Contr}_\pi(\theta).$$

Source-declared incoherent sectors occupy orthogonal summands of $\mathcal{E}_{76}$. An empty active path family gives the defined zero amplitude.

Theorem 5.1 (D2 exact source-amplitude normal form). *Every semantic target-facing D2 transition use of a finite PCR-3/B15 source point factors through $A_{76,a}^{\text{src}}(\theta)$; every summand has a source antecedent; all lawful source arrows act covariantly; and source addition and higher gluing act before physical realization. A missing faithful primitive is SU.*

Proof. Structural induction over the finite source-incidence grammar handles primitive maps, sequential composition, coherent addition, Stinespring projection, constituent derivation, profile integration, and primitive higher-arity contraction. Source coherence makes reduction order immaterial, and every active hyperpath occurs exactly once in its declared coherent or purified sector. The result is an exact transition-amplitude object, not an infinitesimal-generator derivative. $\square$

5.3 Source-defined amplitude space and physical contraction locus

Let $\mathfrak{M}_{fi}^{76}(\mathcal{O}_{76})$ be the target-local off-diagonal linking-algebra corner and retain the fixed kinematic, spectral, symmetry, charge, and owner data. Define the norm-closed complex vector space

$$\begin{aligned} \mathcal{V}_{76}^{\text{src}} &= \{A \in \mathfrak{M}_{fi}^{76}(\mathcal{O}_{76}) : A = P_f^{\text{kin}} A P_i^{\text{kin}}, \\ &\quad E_f(\Delta + \omega_{76})A = A E_i(\Delta), \\ &\quad U_f(g) A U_i(g)^* = \chi_2(g)A, \\ &\quad Q_f A - A Q_i = \Delta q_{76} A, \\ &\quad O_f A = A = A O_i\}. \end{aligned}$$

The physical exact-amplitude locus is

$$\mathcal{A}_{76}^{\text{phys}} = \{A \in \mathcal{V}_{76}^{\text{src}} : \|A\| \leq 1\}.$$

Lemma 5.2 (Amplitude closure and marked physical locus). *Every intrinsic exact amplitude lies in $\mathcal{V}_{76}^{\text{src}}$. The set $\mathcal{A}_{76}^{\text{phys}}$ is closed, balanced, convex, and contains zero. For every $A_H \in \mathcal{V}_{76}^{\text{src}}$, $A_\star \in \mathcal{A}_{76}^{\text{phys}}$, and $z \in [0, 1]$,*

$$C_{H,z} = -A_H + zA_\star \in \mathcal{V}_{76}^{\text{src}}, \quad A_H + C_{H,z} = zA_\star \in \mathcal{A}_{76}^{\text{phys}}.$$

Proof. The typing equations are homogeneous continuous linear equations in a norm-closed linking corner. The contraction-ball statements are standard, and structural recursion preserves the typing equations. $\square$

5.4 Conservative exact-amplitude source extension

Definition 5.3 (Conservative source extension SOE-D2-1). *The extension adds a typed correction-amplitude parameter $C \in \mathcal{V}_{76}^{\text{src}}$, a source presentation $\text{Corr}_{76}(C)$, the colored envelope, ownership, deletion, and exact base reduct. The compiled correction sort is target-relative: it carries D2-only grade and the source grammar contains no constructor from a correction/D2-only input to a frozen oscillation-trace output. A proposal that activates such an old-target incidence is therefore an intrinsic/HYB source presentation (or a target update), not a member of the transparent compiled correction sort. This grading is fixed from the target embedding before response evaluation. The extension does not declare C to be a standalone physical transition. Physicality is tested on the complete amplitude $A_H + C$. No response map, likelihood, acceptance value, or desired endpoint occurs in source formation.*

For a framed base source H , define

$$P_{H,C} = \text{Par}_{76}(H, \text{Own}_{76}(\text{BB}_{76}(H), \text{Corr}_{76}(C))), \quad C \in \mathcal{V}_{76}^{\text{src}},$$

so

$$\mathcal{C}_{\text{CMP}}^{\text{raw}}(H) = \{P_{H,C} : C \in \mathcal{V}_{76}^{\text{src}}\}, \quad A_{H,C}^{\text{tot}} = A_H + C.$$

The reduct forgets the correction and returns H without retuning.

Theorem 5.4 (D2-02: candidate-complete lift universe). *For every B15 frame H , the D2 raw lift universe contains every intrinsic PCR-3 architecture, every compiled point $P_{H,C}$ for $C \in \mathcal{V}_{76}^{\text{src}}$, and every admitted finite hybrid with higher incidence retained. Formation is independent of response and endpoint verdicts; each compiled point has exact reduct H .*

Proof. Intrinsic coverage is PCR-3 semantic completeness specialized by the amplitude normal form. The compiled component is the complete C -indexed family generated by the declared extension. Every raw source point exists before physicality and response evaluation, so response mutation alters only the downstream status partition. $\square$

The extension generates typed correction presentations, not physical occupants by AASC fiat. A physical process is constructed only for the complete amplitude of a candidate that passes the contraction and target-typing conditions.

6 Source-authorized exact-amplitude completion

6.1 Exact physical amplitude category

For $A \in \mathcal{A}_{76}^{\text{phys}}$, define

$$D_A = (I - A^*A)^{1/2}, \quad D_{A^*} = (I - AA^*)^{1/2},$$

and

$$\mathbb{U}_{76}(A) = \begin{pmatrix} A & D_{A^*} \\ D_A & -A^* \end{pmatrix} : \mathcal{H}_i^{76} \oplus (\mathcal{H}_f^{76} \otimes \mathcal{E}_{76}) \rightarrow (\mathcal{H}_f^{76} \otimes \mathcal{E}_{76}) \oplus \mathcal{H}_i^{76}.$$

Theorem 6.1 (Exact transition-amplitude realization). *For every $A \in \mathcal{A}_{76}^{\text{phys}}$, $\mathbb{U}_{76}(A)$ is unitary and*

$$P_f \mathbb{U}_{76}(A) P_i = A.$$

The object $\text{RealAmp}_{76}(A) = (A, \mathbb{U}_{76}(A), \text{Own}_{76})$ satisfies the applicable support, kinematic, covariance, charge-transfer, owner, and exact-transition residuals and is natural under lawful arrows.

Proof. The defect identity $AD_A = D_{A^*}A$ proves unitarity. The upper-left block is literally A . Since A intertwines the frozen spectral, symmetry, charge, and owner data, A^*A and AA^* commute with the corresponding representations; functional calculus transports those properties to the defect operators. Local linking-algebra membership places the diagonal defect operators in the associated local diagonal algebras. No infinitesimal derivative is used. The defect blocks are the D2-only complementary no-transition/completion channels required by unitarity. They lie in the corresponding diagonal target algebras, are invisible to the admitted transition projection $P_f(\cdot)P_i$, and are removed with the enriched exact-amplitude presentation by the base reduct. They therefore do not alter the retained oscillation packet. $\square$

6.2 Split-anchored physical diagrams, semantic residuals, and activity reduction

For each framed B15 source H , choose a semantic PCR-4/B15 physical representative W_H of its standing core. No canonical representative is claimed. Let $\mathcal{B}_{76}(W_H)$ denote the complete retained attachment diagram: the charged-current source and detector faces, owner modules, representation and charge anchors, matching interfaces, vacuum branch, retained continuation data, and the concrete PCR-4 common-witness maps on which an enriched process must agree with the base realization.

Define $\text{Phys}_{D2}^{\text{anc}}$ to be the category of finite *split-anchored* Hilbert/operator witness diagrams. An object over W_H consists of a finite D2 cell diagram $\mathcal{D}_{76}(H, C)$, a closed isometric attaching map from its boundary $\partial\mathcal{D}_{76}(H, C)$ into $\mathcal{B}_{76}(W_H)$, and a specified orthogonal splitting of every process sort into its attached boundary part and D2-only complement. Operators, channels, source-coloured incidence maps, owner and matching arrows, continuation maps, and higher relations must agree on the attached boundary. Morphisms are componentwise lawful isometries or physical equivalences preserving the attaching maps, splittings, and every typed relation.

The semantic simultaneous physical locus is the full subcategory

$$\text{PhysMod}_{4,D2}^{\text{sem}} \subset \text{Phys}_{D2}^{\text{anc}}$$

whose objects satisfy the complete active PCR-4/D2 physical residual packet. The accepted semantic survivor category is

$$\text{CarrierSurv}_{\text{sem}}(\hat{T}_{D2}^{\text{Ge}}) = \{(W, \eta) : W \in \text{PhysMod}_{4,D2}^{\text{sem}}, \eta \in \mathcal{E}_{76}^{\text{adm}}, E_{\text{resp}}(W, \eta) = 0, \text{Stat}(W, \eta) = \text{Pass}\}.$$

Thus physical amalgamation and empirical acceptance are separate stages. A physical object may belong to $\text{PhysMod}_{4,D2}^{\text{sem}}$ while a particular response frame fails or is out of scope; it enters $\text{CarrierSurv}_{\text{sem}}(\hat{T}_{D2}^{\text{Ge}})$ only in one lawful complete frame whose response is defined and accepted.

The source extension makes correction activity pointwise and response blind. Put

$$\alpha_{76}(C) = \begin{cases} 0, & C = 0, \\ 1, & C \neq 0, \end{cases} \quad P_{H,C}^{\text{red}} := \text{ActRed}(P_{H,C}).$$

Because every owner, attachment, and derived incidence of $\text{Corr}_{76}(C)$ is generated from the correction amplitude, the zero correction is source-null. Hence

$$P_{H,C}^{\text{red}} \simeq \begin{cases} H, & C = 0, \\ P_{H,C}, & C \neq 0. \end{cases} \quad (7)$$

This is a source equation fixed before response evaluation. A future proposal assigning independent constitutive activity to $\text{Corr}_{76}(0)$ is a source update rather than a reinterpretation of the present trace.

6.2.1 Complete residual partition

Let $\mathcal{K}_{4,D2}^{\text{act}}(H, C)$ be the active simultaneous relation family for the compiled point. Its physical residual packet is partitioned disjointly as

$$\mathcal{E}_{4,D2}^{\text{phys}} = (\mathcal{E}_{\text{inh}}, \mathcal{E}_{\text{proc}}, \mathcal{E}_{\text{cross}}, E_{\text{sim}}).$$

The partition is by the minimal support of the relation, not by a preferred ordering.

Residual band	Exact content
Inherited	Every active source, target, nested-role, charged-current, retained oscillation-channel/response, base locality/environment, base gauge/representation/charge/anomaly, base vacuum/deformation, base EFT/extension, base relation, no-repair, and equivalence residual whose support lies entirely in the retained B15 diagram.
Pure D2 process	Exact-amplitude contraction, support, kinematic spectral shift, covariance, target-fixed $\Delta L = 2$ charge transfer, owner typing, unitarity, and exact transition-block residuals of the D2 process leg.
Cross	Boundary and owner attachment, source-to-target matching, continuation, shared vacuum/deformation branch, causal overlap, induced-effect compatibility, mixed source relations, and every active primitive or higher incidence whose support meets both the retained and D2-only cells.
Common cone	Recovery of every active witness by restriction or projection from one object, with all overlap diagrams commuting inside the semantic model category.

For the exact total amplitude $A = A_H + C$, the pure process residual is represented explicitly by

$$\begin{aligned} \mathcal{E}_{\text{proc}}(A) := & (A - P_f \mathbb{U}_{76}(A) P_i, \mathbb{U}_{76}(A)^* \mathbb{U}_{76}(A) - I, \mathbb{U}_{76}(A) \mathbb{U}_{76}(A)^* - I, \\ & A - P_f^{\mathcal{O}} A P_i^{\mathcal{O}}, S_f A - A S_i - \Delta_{76} A, \rho_f(g) A - A \rho_i(g), \\ & Q_f A - A Q_i - \Delta Q_{76} A, E_{\text{own}}(A))_{g \in G_{76}}. \end{aligned} \quad (8)$$

Membership $A \in \mathcal{A}_{76}^{\text{phys}}$ means precisely that the support, spectral, covariance, charge-transfer, and owner coordinates vanish; [theorem 6.1](#) gives the three colligation coordinates.

Write ι_s for the base inclusion, κ_s for the D2-complement inclusion, and r_s for the split retraction at a physical sort s .

Lemma 6.2 (Canonical physical relative-correction datum). *For every framed semantic base realization W_H and every $C \in \mathcal{V}_{76}^{\text{src}}$ with $A_H + C \in \mathcal{A}_{76}^{\text{phys}}$, there is a source-faithful relative D2 datum*

$$\mathfrak{D}_{76}^{\text{rel}}(H, C) = (M_C, U_C, \Omega_C, D_C, I_C, I_{\times}, \{\mu_e^C, \mu_e^{\times}\}_{e \in E_{76}^{\text{act}}})$$

formed before response evaluation, with the following properties.

1. The correction process is a morphism between the already fixed boundary modules. It introduces no new primitive chiral representation, gauge charge lattice, anomaly-bearing field, or independent vacuum branch. Hence

$$\mathcal{A}_{\text{anom}}(\mathfrak{D}_{76}^{\text{rel}}) = 0, \quad \Omega_C = 0, \quad D_C(t) = I$$

on the D2-only process complement, while gauge covariance and charge transfer are exactly the intertwining equations contained in $A_H + C \in \mathcal{A}_{76}^{\text{phys}}$.

2. Matching and continuation on the complement are the maps induced by the exact-amplitude colligation and the licensed target continuation. On the pre-pushout direct sum they are the block maps

$$M_H \oplus M_C, \quad U_H^{s \rightarrow t} \oplus U_C^{s \rightarrow t}, \quad D_H(t) \oplus I.$$

3. Every mixed induced-effect or higher-incidence map is either the zero map when no mixed source cell is declared, or the bounded operator obtained by interpreting the corresponding finite source hyperpath of $P_{H,C}$. Its face, composition, adjoint, and ownership identities are the physically represented constitutive equations of that source cell.
4. The datum is natural under lawful base and correction arrows and is unchanged by any mutation of nuclear, detector, statistical, likelihood, or acceptance data.

Proof. The exact-amplitude typing equations make $A_H + C$ an intertwiner of the fixed support, spectral, symmetry, charge, and owner modules. The defect operators are obtained by functional calculus from $(A_H + C)^*(A_H + C)$ and $(A_H + C)(A_H + C)^*$, so the same intertwiners extend to the colligation. Since the correction sort is an effective process morphism rather than a new primitive chiral carrier, its anomaly contribution and independent vacuum/deformation load are zero by the frozen SOE-D2-1 typing. Block extension then gives matching, continuation, owner, vacuum, and deformation maps that agree literally with the base maps on the common anchor. The finite source-incidence interpretation theorem supplies each nonzero mixed or higher map and its constitutive identities; undeclared mixed cells are represented by zero. All constructions use only the base frame, C , and the frozen source/physical typing, proving response blindness and naturality. $\square$

The relative correction datum therefore contains maps

$$M_C, \quad U_C, \quad \Omega_C, \quad D_C(t), \quad I_C, \quad I_{\times}(H, C), \quad \{\mu_e^C, \mu_e^{\times}\}_{e \in E_{76}^{\text{act}}},$$

obtained by interpreting the source cells of $\text{BB}_{76}(H)$ and $\text{Corr}_{76}(C)$ and by applying the exact-amplitude colligation on the D2 process sorts. By [lemma 6.2](#), they obey the following cross equations before the pushout is formed:

$$M_{H,C}\iota_s = \iota_t M_H, \quad M_{H,C}\kappa_s = \kappa_t M_C, \quad (9)$$

$$U_{H,C}^{s \rightarrow t} \iota_s = \iota_t U_H^{s \rightarrow t}, \quad U_{H,C}^{s \rightarrow t} \kappa_s = \kappa_t U_C^{s \rightarrow t}, \quad (10)$$

$$r\Omega_{H,C} = \Omega_H, \quad \Omega_{H,C} = \iota\Omega_H + \kappa\Omega_C, \quad (11)$$

$$D_{H,C}(t)\iota = \iota D_H(t), \quad D_{H,C}(t)\kappa = \kappa D_C(t), \quad (12)$$

$$\text{Own}_{H,C} \iota = \iota \text{Own}_H, \quad \text{Own}_{H,C} \kappa = \iota_{\text{own}} \text{Own}_C. \quad (13)$$

The causal/support relation is

$$\text{supp}(A_H + C) \subseteq \mathcal{O}_{76}, \quad [\mathfrak{A}_H(\mathcal{O}_1), \mathfrak{A}_C(\mathcal{O}_2)] = 0 \quad \text{for } \mathcal{O}_1 \perp \mathcal{O}_2 \quad (14)$$

outside the identified anchor. The induced-effect relation is the exact source-interpreted decomposition

$$I_{H,C} = I_H + I_C + I_{\times}(H, C), \quad \Pi_{76}^{\text{tr}}(I_{H,C}) = A_H + C, \quad (15)$$

where $I_{\times}$ is the finite sum of the mixed source hyperpaths already present in the compiled source. For every active hyperedge $e : s_1, \dots, s_m \rightarrow t$, the primitive and higher incidence equations are

$$\begin{aligned} \mu_e^{H,C} \circ \bigotimes_{j=1}^m \iota_{s_j} &= \iota_t \circ \mu_e^H, \\ \mu_e^{H,C} \circ \bigotimes_{j=1}^m \kappa_{s_j} &= \kappa_t \circ \mu_e^C, \\ d_r \mu_e^{H,C} &= \mu_{d_{re}}^{H,C}, \quad \mu_e^{H,C} \circ \mu_f^{H,C} = \mu_{e \circ f}^{H,C}, \end{aligned} \quad (16)$$

with the mixed-input faces supplied by $\mu_e^{\times}$. Equations (9)–(16) are physical equalities in the relative correction datum; they are not inferred from the mere presence of similarly named arrows.

6.2.2 Concrete split attachment

For $C \in \mathcal{V}_{76}^{\text{src}}$ with $A_H + C \in \mathcal{A}_{76}^{\text{phys}}$, construct $\text{D}_{76}(H, C)$ source-first by adjoining to the existing cells of W_H :

1. the coloured hyperprocess decorations of $\text{BB}_{76}(H)$, attached to the existing base cells rather than copied as a second base realization;
2. the correction cell $\text{Corr}_{76}(C)$, its owner relation, and every declared D2 primitive or higher incidence;
3. the constitutive amplitude cell $A_{H,C}^{\text{tot}} = A_H + C$;
4. the exact unitary colligation $\mathbb{U}_{76}(A_H + C)$, its D2-only defect channels, and its transition projection;
5. the maps in (9)–(16), together with every D2 failure and response-interface arrow generated by the compiled source point.

All attaching data are fixed before response evaluation. At each Hilbert sort s , the physical cell attachment is the concrete split pushout

$$\mathcal{H}_s(W_{H,C}) = \frac{\mathcal{H}_s(W_H) \oplus \mathcal{H}_s(\text{D}_{76}(H, C))}{\langle \iota_H^s \xi - \iota_D^s \xi : \xi \in \mathcal{H}_s(\partial \text{D}_{76}(H, C)) \rangle} \simeq \mathcal{H}_s(W_H) \oplus \mathcal{K}_s(H, C). \quad (17)$$

The resulting decorated diagram is

$$W_{H,C} := W_H \bigcup_{\mathbf{B}_{76}(W_H)}^{\text{phys}} \mathbf{D}_{76}(H, C).$$

The split form supplies a faithful base inclusion $\iota_H : W_H \hookrightarrow W_{H,C}$ and a physical retraction $r_H : W_{H,C} \rightarrow W_H$ that is the identity on the complete base diagram and removes only D2-only cells.

Lemma 6.3 (Split residual decomposition). *For every active residual $E_j \in \mathcal{K}_{4,D2}^{\text{act}}(H, C)$, evaluation on the split attachment has exactly one of the forms*

$$E_j(W_{H,C}) = \begin{cases} \iota E_j(W_H), & j \in \mathcal{K}_{\text{inh}}, \\ \kappa E_j^{\text{proc}}(A_H + C), & j \in \mathcal{K}_{\text{proc}}, \\ E_j^{\text{cross}}(H, C), & j \in \mathcal{K}_{\text{cross}}, \\ E_{\text{sim}}(W_{H,C}), & j = \text{sim}. \end{cases}$$

The first three lines vanish, and E_{sim} vanishes with common apex $W_{H,C}$.

Proof. Every residual is an equality of finite composites, restrictions, tensor contractions, adjoints, functional-calculus maps, or finite higher-incidence operations in the PCR-4/D2 relation theory. Under the canonical decomposition in (17), its evaluation has base, D2-process, and mixed blocks. The base block is zero because W_H is a semantic PCR-4 survivor. The pure D2 block is zero by $A_H + C \in \mathcal{A}_{76}^{\text{phys}}$ and theorem 6.1. The mixed blocks are exactly (9)–(16); hence they vanish. In particular, lemma 6.2 gives zero new anomaly load and no independent vacuum branch, while the exact-amplitude intertwining equations close the gauge and charge band. Thus no active cross residual is discharged merely because a named arrow is present: its displayed equation is zero on the pushout. All active witnesses and all their overlap maps are restrictions or projections of the same zero-residual quotient object, so that object is a common cone inside the semantic zero locus rather than merely a cone in the ambient diagram category. $\square$

Theorem 6.4 (Constraint-Preserving Physical Amalgamation and Source-Faithful Enriched Realization). *For every framed PCR-5B/B15 physical realization W_H and every compiled source point $P_{H,C}$ with $A_H + C \in \mathcal{A}_{76}^{\text{phys}}$, the split attachment functor*

$$\mathfrak{A}_{76} : (W_H, C) \mapsto W_{H,C}$$

lands in the semantic simultaneous physical model locus $\text{PhysMod}_{4,D2}^{\text{sem}}$. More precisely:

1. its activity-reduced carrier trace is

$$\chi_{\text{car}}(W_{H,C}) \simeq P_{H,C}^{\text{red}};$$

2. its distinguished D2 transition projection is exact,

$$\Pi_{76}^{\text{tr}}(W_{H,C}) = P_f \mathbb{U}_{76}(A_H + C) P_i = A_H + C;$$

3. its physical oscillation restriction is the original realization,

$$\text{Res}_{\text{osc}}(W_{H,C}) := r_H(W_{H,C}) \simeq W_H;$$

4. every inherited, pure-process, and cross residual vanishes by the classwise calculation of lemma 6.3;

5. the complete witness diagram has a common cone inside $\text{PhysMod}_{4,D2}^{\text{sem}}$, with apex $W_{H,C}$;

6. *physical correction deletion is the split retraction and is natural and no-retuning,*

$$\text{Del}_{c_{76}}^{\text{phys}}(W_{H,C}) = r_H(W_{H,C}) \simeq W_H;$$

7. *the source reduct is the carrier-trace shadow of physical restriction: the square*

$$\begin{array}{ccc} W_{H,C} & \xrightarrow{\chi_{\text{car}}} & P_{H,C}^{\text{red}} \\ \downarrow \text{Res}_{\text{osc}} = \text{Del}_{c_{76}}^{\text{phys}} & & \downarrow U_{D2}^{\text{CMP}} \\ W_H & \xrightarrow{\chi_{\text{car}}} & H \end{array}$$

commutes up to the lawful B15 framing isomorphism.

The assignment is natural under lawful base and correction arrows. For a complete response frame η , the pair $(W_{H,C}, \eta)$ lies in $\text{CarrierSurv}_{\text{sem}}(\widehat{T}_{D2}^{\text{Ge}})$ exactly on the defined accepted response locus.

Proof. The construction of $D_{76}(H, C)$ gives a relative correction model with the explicit equations (8) and (9)–(16). The split pushout exists by (17). The split residual decomposition proves that every active PCR-4/D2 physical residual vanishes and that the common apex lies in the semantic model locus. This proves items (2)–(5).

The base cells occur once through ι_H . If $C \neq 0$, the correction cell and its derived incidences are source-active and the activity-reduced trace is $P_{H,C}$; if $C = 0$, they are source-null and are stripped, giving H . This is exactly (7), proving (1). Removing the D2-only complement is r_H , which fixes every base operator, parameter, incidence, and witness and cannot recompute C , proving (6). Applying χ_{car} to that physical retraction gives the exact source reduct of the activity-reduced trace, proving (7). Lawful arrows preserve the relative equations and induce arrows of the split pushouts, giving naturality. The final response statement follows from the separate fibred response and status compiler. $\square$

6.3 Retained-oscillation transparency and base conservativity

Physical retraction and correction deletion are not the definition of target restriction. Let

$$\Phi_4^{\text{ret}} : \text{PhysMod}_{4,D2}^{\text{sem}} \longrightarrow \mathcal{P}_4^{\text{osc}}, \quad \Xi_{\text{osc}}^{\text{ret}} : \text{PhysMod}_{4,D2}^{\text{sem}} \longrightarrow \mathcal{X}_{\text{osc}}$$

be the *in-situ retained oscillation evaluators*: the fixed PCR-4 evaluator grammar and its seven-channel SCIC projection are applied to the enriched physical object with every D2 cell still present, and only the old-target output sorts are read. In particular,

$$\Phi_4^{\text{ret}}(W) \text{ is not defined as } \Phi_4(\text{Del}_{c_{76}}^{\text{phys}} W),$$

and no physical correction deletion is permitted in a proof of retained equality.

For the compiled source signature, let $\mathcal{I}_{76}^{D2}(H, C)$ be the closed target-relative ideal generated by the correction cell, exact-amplitude completion/defect cells, D2-only nuclear-transition attachment, D2 response/failure cells, and every higher source cell whose output sort is D2-only. The fixed SOE-D2-1 typing has two decisive properties:

1. every old-target source, propagation, charged-current boundary, matching, continuation, deletion, and failure-output sort belongs to the retained grade carried by W_H ;
2. there is no compiled-chart constructor with at least one $\mathcal{I}_{76}^{D2}$ -input and output in a fixed oscillation-trace sort. A source proposal containing such an incidence is generated as an intrinsic/HYB extension and is re-evaluated at its actual retained packet; it is not silently retained in the transparent compiled chart.

These conditions are source/target typing fixed before D2 response evaluation. They do not depend on the measured likelihood, on A_{76}^* , or on a desired base class.

Lemma 6.5 (Frozen oscillation evaluator annihilation). *For every primitive generator e of the frozen PCR-4/SCIC oscillation evaluator algebra and every licensed old-target continuation u , the in-situ extension to a compiled physical point satisfies*

$$e^{\text{ret}}(u(W_{H,C})) = e(u(W_H)), \quad e^{\text{ret}} \upharpoonright_{\mathcal{I}_{76}^{D2}(H,C)} = 0,$$

whenever $A_H + C \in \mathcal{A}_{76}^{\text{phys}}$. Every mixed evaluator term with at least one D2-only input and an old-target output vanishes.

Proof. At every split sort, (17) gives the retained/D2 decomposition $\mathcal{H}_s(W_{H,C}) \simeq \mathcal{H}_s(W_H) \oplus \mathcal{K}_s(H, C)$. The old-target generators act on the first summand exactly as on W_H . By the relative-grade formation rule, their old-target output projection is zero on $\mathcal{K}_s(H, C)$, and no mixed source cell of the compiled chart has an old-target output. The block continuation relations (10) preserve this grading. Structural induction over composition, tensor contraction, adjoint, functional calculus, finite higher incidence, and licensed continuation therefore preserves the annihilation property. The enriched object is never replaced or deleted during this evaluation; the result is factorization of the old-target evaluator itself through the retained grade. $\square$

Write the fixed seven-channel packet as

$$\Xi_{\text{osc}} = (T^{\text{face}}, R^{\text{stand}}, \Delta_{\text{fr}}^{\text{del}}, C_{\text{act}}, M_{\text{match}}, B_{\text{bdy}}, F_{\text{same}}).$$

Theorem 6.6 (Retained-Oscillation Transparency and Base-Conservativity). *For every framed PCR-5B/B15 physical realization W_H and every compiled physical point $W_{H,C}$ with $A_H + C \in \mathcal{A}_{76}^{\text{phys}}$, the complete frozen oscillation target is conserved while the correction remains physically present:*

$$\boxed{\Phi_4^{\text{ret}}(W_{H,C}) = \Phi_4(W_H)}, \quad \boxed{\Xi_{\text{osc}}^{\text{ret}}(W_{H,C}) = \Xi_{\text{osc}}(W_H)}.$$

Channel by channel,

$$\begin{aligned} T^{\text{face}, \text{ret}} &= T^{\text{face}}, & R^{\text{stand}, \text{ret}} &= R^{\text{stand}}, & \Delta_{\text{fr}}^{\text{del}, \text{ret}} &= \Delta_{\text{fr}}^{\text{del}}, \\ C_{\text{act}}^{\text{ret}} &= C_{\text{act}}, & M_{\text{match}}^{\text{ret}} &= M_{\text{match}}, & B_{\text{bdy}}^{\text{ret}} &= B_{\text{bdy}}, & F_{\text{same}}^{\text{ret}} &= F_{\text{same}}. \end{aligned} \quad (18)$$

No equality in this theorem is obtained by applying $\text{Del}_{c76}^{\text{phys}}$, r_H , or by retuning C .

Define the actual transparent physical locus

$$\mathcal{V}_{76}^{\text{tr}}(H) := \left\{ C \in \mathcal{V}_{76}^{\text{src}} : A_H + C \in \mathcal{A}_{76}^{\text{phys}}, \Xi_{\text{osc}}^{\text{ret}}(W_{H,C}) = \Xi_{\text{osc}}(W_H) \right\}.$$

For the compiled SOE-D2-1 chart,

$$\boxed{\mathcal{V}_{76}^{\text{tr}}(H) = \{ C \in \mathcal{V}_{76}^{\text{src}} : A_H + C \in \mathcal{A}_{76}^{\text{phys}} \}}.$$

Proof. The full PCR-4 equality follows from lemma 6.5 by saturation of the frozen target-evaluator algebra. For the packet coordinates, the factorization has the following concrete consequences.

1. T^{face} : the compiled correction has no old charged-current source/detection face; the D2 nuclear transition face is an enriched-only sort.
2. R^{stand} : every licensed old-target propagation/standing continuation preserves the retained/D2 grading, so its old-target output is exactly the base continuation output.

3. $\Delta_{\text{fr}}^{\text{del}}$: each frozen base no-repair deletion is evaluated on the enriched object with C held fixed. The correction cannot repair or alter an old-target deletion consequence because the relative ideal is annihilated by the old evaluator. Thus the deletion signature equals the base signature without deleting or recomputing C .
4. C_{act} : activity for the frozen oscillation target is computed from retained-grade incidences. The correction is D2-only and does not enter the old active-kind set.
5. M_{match} : the old matching map is the retained block of (9); every D2 matching output is enriched-only and has zero old-target projection.
6. B_{bdy} : the correction attaches to the D2 nuclear-transition boundary and does not alter the frozen charged-current oscillation boundary.
7. F_{same} : old-target failure and same-source continuation outputs factor through the retained grade under every licensed continuation; D2-only failure cells remain enriched coordinates and do not enter the old failure output.

Hence all seven coordinates agree, and the same induction gives the complete Φ_4 equality. Since the argument uses only the compiled relative-grade typing and physicality of the complete amplitude, every physically admissible compiled correction lies in $\mathcal{V}_{76}^{\text{tr}}(H)$; the reverse inclusion is definitional. $\square$

Corollary 6.7 (Transparent marked family and routing of nontransparent proposals). *For every framed base realization and every $z \in [0, 1]$,*

$$C_{H,z}^* \in \mathcal{V}_{76}^{\text{tr}}(H) \quad \text{whenever} \quad zA_{76}^* \in \mathcal{A}_{76}^{\text{phys}}.$$

In particular, every accepted marked point used below lies over the same frozen base standing core before correction deletion.

A source proposal that activates an old-target incidence is not classified as a failed transparent correction. It is routed by its in-situ retained packet: if that packet is a valid PCR-5B standing core, $\text{Route}_{\text{osc}}$ assigns the corresponding base class; if it changes the old target/interface rather than merely its standing core, the point receives TU. Such proposals remain in the intrinsic/HYB candidate universe and are not omitted from D2-02.

Corollary 6.8 (Complete-amplitude physicality). *A raw compiled point $P_{H,C}$ has a source-faithful simultaneous physical realization precisely where $A_H + C \in \mathcal{A}_{76}^{\text{phys}}$. There the full object is $W_{H,C} \in \text{PhysMod}_{4,D2}^{\text{sem}}$, its D2 process projection is $\text{RealAmp}_{76}(A_H + C)$, its activity-reduced trace is $P_{H,C}^{\text{red}}$, and its physical base restriction is W_H . The correction C need not be a standalone contraction.*

Theorem 6.9 (Exact complete-amplitude compatibility with an arbitrary base realization). *For every framed PCR-5B/B15 physical realization W_H , every source-generated correction coordinate $C \in \mathcal{V}_{76}^{\text{src}}$, and every lawful complete response frame η , the constraint-preserving object $W_{H,C}$ has exact additive target-facing amplitude*

$$\mathcal{A}_{76}^{\text{tot}}(W_{H,C}) = A_H + C.$$

Its D2 process projection is $\text{RealAmp}_{76}(A_H + C)$, its physical base restriction is W_H , and its exact event kernel is

$$K_{H,C,\eta} = \mathcal{R}_{76,\eta}(A_H + C) = \mathcal{R}_{76,\eta}(A_H) + \mathcal{R}_{76,\eta}(C).$$

The exact rate is

$$\Gamma_{1/2}^{0\nu}(W_{H,C}, \eta) = \mathfrak{G}_{76,\eta}(K_H, K_H) + 2 \text{Re } \mathfrak{G}_{76,\eta}(K_H, K_C) + \mathfrak{G}_{76,\eta}(K_C, K_C),$$

where $K_H = \mathcal{R}_{76,\eta}(A_H)$ and $K_C = \mathcal{R}_{76,\eta}(C)$. Every term is evaluated on the same semantic physical apex $W_{H,C}$.

Proof. The source equation $A_{H,C}^{\text{tot}} = A_H + C$ is faithfully represented in the process leg of $W_{H,C}$. The constraint-preserving amalgamation theorem proves that this leg and the retained base realization form one semantic physical model rather than merely one ambient diagram. Projection to the distinguished colligation gives the exact finite amplitude. Linearity of the fibred amplitude-to-record map and Hermiticity of the rate form give the displayed kernel and interference expansion. Physical restriction and deletion preserve the original base realization without retuning. $\square$

6.4 Response-visible exact amplitude

After the raw source domain is complete, choose a source-authorized exact amplitude and lawful response frame

$$(A_{76}^*, \eta_*) \in \mathcal{A}_{76}^{\text{phys}} \times \mathcal{E}_{76}^{\text{adm}}$$

with $K_{*,\eta_*} = \mathcal{R}_{76,\eta_*}(A_{76}^*) \neq 0$ and $\gamma_{*,\eta_*} > 0$.

Lemma 6.10 (Response-visible exact-amplitude witness). *Such a pair exists before the observed LEGEND likelihood bound is applied.*

Proof. The complete raw source and response domains have already been formed. Within them choose, solely for this downstream nondegeneracy proof, one source-authorized LAW/LME amplitude $A_0 \in \mathcal{V}_{76}^{\text{src}}$ and one lawful complete ^{76}Ge response frame containing a nonzero nuclear-transition representative, normalized signal instrument, and statistical response object. Chiral EFT, including the leading short-range contribution, maps this already-generated source amplitude to a nonzero two-electron transition response [16–18, 21]; the frozen nuclear package contains explicit nonzero ^{76}Ge response representatives [20]. Scale only the already-generated source parameter by $0 < \lambda \leq \min\{1, \|A_0\|^{-1}\}$. Then $A_{76}^* = \lambda A_0 \in \mathcal{A}_{76}^{\text{phys}}$, its response remains nonzero, and positivity of the Hermitian form is strict on that response-visible kernel. The resulting pair is a marked witness inside already complete source and response domains; it is not an argument of source formation and is fixed before the observed likelihood bound is applied. $\square$

6.5 Marked exact-amplitude survivor family

For each framed base source, put $A_H = \mathcal{A}_{76}^{\text{src}}(H)$ and define, only after the witness pair exists,

$$C_{H,z}^* = -A_H + zA_{76}^*, \quad P_{H,z}^* = P_{H,C_{H,z}^*}, \quad P_{H,z}^{*,\text{red}} := \text{ActRed}(P_{H,z}^*) = P_{H,C_{H,z}^*}^{\text{red}}, \quad z \in [0, 1].$$

Then

$$A_{H,z}^{\text{tot}} = A_H + C_{H,z}^* = zA_{76}^* \in \mathcal{A}_{76}^{\text{phys}}.$$

Theorem 6.11 (Marked source-faithful affine-survivor theorem). *For every framed base realization W_H , $z \in [0, 1]$, and lawful complete response frame η , define*

$$W_{H,z}^* := W_{H,C_{H,z}^*}.$$

Then

$$\chi_{\text{car}}(W_{H,z}^*) \simeq P_{H,z}^{*,\text{red}}, \quad \text{Res}_{\text{osc}}(W_{H,z}^*) \simeq W_H,$$

and its distinguished D2 process projection has exact transition block zA_{76}^* . Its event kernel and complete rate are

$$K_{H,z,\eta}^{\text{tot}} = zK_{*,\eta}, \quad \Gamma_{1/2}^{0\nu}(W_{H,z}^*, \eta) = z^2\gamma_{*,\eta}.$$

The full intrinsic, correction, and interference decomposition is retained on the same physical object before simplification. Physical correction deletion satisfies

$$\text{Del}_{c_{76}}^{\text{phys}}(W_{H,z}^*) \simeq W_H.$$

Proof. The exact source-amplitude identity holds before realization. The Constraint-Preserving Physical Amalgamation and Source-Faithful Enriched Realization Theorem constructs one physical apex realizing the compiled source and retaining W_H . The general compatibility theorem retains intrinsic, correction, and interference contributions on that apex. Since $zA_{76}^* \in \mathcal{A}_{76}^{\text{phys}}$, the distinguished process projection has literal block zA_{76}^* . The fibred response is linear on exact amplitudes and the rate form is Hermitian. Deletion naturality is part of [theorem 6.4](#). $\square$

At η_* , define

$$I_{76,\eta_*}^{\text{acc}} = \{z \in [0, 1] : \text{Acc}_{90,\eta_*}(zK_{*,\eta_*}) = 1\} = \{z \in [0, 1] : z^2\gamma_{*,\eta_*} < \Gamma_{90}^{\text{Ge}}\},$$

and put

$$\delta_{76}^* = \frac{1}{2} \min \left\{ 1, \sqrt{\Gamma_{90}^{\text{Ge}}/\gamma_{*,\eta_*}} \right\} > 0, \quad I_{76,\eta_*}^* = [0, \delta_{76}^*] \subset I_{76,\eta_*}^{\text{acc}}.$$

The first set is the exact marked pass locus at η_* ; the second is only the continuum-cardinality witness. For each base frame define

$$Z_H^{\text{null}} := \{z \in I_{76,\eta_*}^* : C_{H,z}^* = 0\}, \quad I_{76,H}^{*,\text{act}} := I_{76,\eta_*}^* \setminus Z_H^{\text{null}}.$$

Because $A_{76}^* \neq 0$, the equation $-A_H + zA_{76}^* = 0$ has at most one solution. Hence

$$|Z_H^{\text{null}}| \leq 1, \quad |I_{76,H}^{*,\text{act}}| = \mathfrak{c}.$$

The active interval is used whenever the proof requires a nonnull correction block; the exact survivor locus continues to include correctly activity-reduced zero-correction points.

Lemma 6.12 (Applicable residual inheritance). *For every $z \in I_{76,\eta_*}^{\text{acc}}$, the marked source, exact-amplitude realization, and complete response frame η_* satisfy every applicable enriched residual. All intrinsic local-QFT induced terms occur in A_H before cancellation.*

Proof. The constraint-preserving amalgamation theorem gives $W_{H,z}^* \in \text{PhysMod}_{4,D2}^{\text{sem}}$ because its exact total amplitude is $zA_{76}^* \in \mathcal{A}_{76}^{\text{phys}}$. By [lemma 6.3](#), inherited residuals equal the already vanishing base residuals, pure-process residuals vanish by exact-amplitude physicality, every matching, owner, continuation, vacuum/deformation, causal, induced-effect, and higher-incidence cross residual vanishes by (9)–(16), and the common apex lies inside the semantic model locus. Independently, [theorem 6.6](#) and [corollary 6.7](#) prove that the complete frozen oscillation evaluator and all seven SCIC channels are unchanged while the correction remains present. The response at η_* is defined and accepted for $z \in I_{76,\eta_*}^{\text{acc}}$, so $(W_{H,z}^*, \eta_*) \in \text{CarrierSurv}_{\text{sem}}(\hat{T}_{D2}^{\text{Ge}})$. No response frame is spliced and no correction is retuned. $\square$

6.6 Local-QFT black-box guardrail

In local $SU(2)_L \times U(1)_Y$ settings with the usual charged-current structure, a nonzero $0\nu\beta\beta$ amplitude can induce a Majorana self-energy [22, 23]. The induced mass may be extremely small, and cancellation constructions block a unique microscopic-character theorem without additional hypotheses [24, 25]. Every intrinsic local-QFT chart therefore retains all induced self-energy, matching, and continuation consequences in A_H . The target-scale correction process adds a separately realized exact amplitude; it does not erase the intrinsic source history or change the base reduct.

7 One-point simultaneous D2 survival

7.1 Complete enriched witness packet

For a raw point (X, η) , let

$$\Phi_{D2}(X, \eta) = (\Phi_4^{\text{ret}}, \Xi_{\text{osc}}^{\text{ret}}, [A_X]_{D2}^{\text{St}}, K_{76, \eta}, \Gamma_{1/2}^{0\nu}(\eta), \mathcal{I}_{76, \eta}, A^{D2}, \text{Stat}^{D2}, \text{Comp}^{D2}, \text{Prov}^{D2})$$

record retained base data, standing-amplitude class, response, status, composition, and provenance.

Theorem 7.1 (D2-03: constraint-preserving source-faithful one-point oscillation plus LNV survivor theorem). *For every framed B15 physical base realization W_H and every $z \in I_{76, \eta_\star}^{\text{acc}}$, the one-point enriched object*

$$X_{H,z} := (W_{H,z}^\star, \eta_\star)$$

is a D2 survivor. It satisfies

$$\chi_{\text{car}}(W_{H,z}^\star) \simeq P_{H,z}^{\star, \text{red}}, \quad \text{Res}_{\text{osc}}(W_{H,z}^\star) \simeq W_H,$$

the complete frozen oscillation evaluation is conserved in situ,

$$\Phi_4^{\text{ret}}(W_{H,z}^\star) = \Phi_4(W_H), \quad \Xi_{\text{osc}}^{\text{ret}}(W_{H,z}^\star) = \Xi_{\text{osc}}(W_H),$$

its exact complete transition amplitude and D2 process projection are

$$A_{H,z}^{\text{tot}} = zA_{76}^\star, \quad \Pi_{76}^{\text{tr}}(W_{H,z}^\star) = zA_{76}^\star,$$

and

$$K_{H,z, \eta_\star}^{\text{tot}} = zK_{\star, \eta_\star}, \quad \Gamma_{1/2}^{0\nu}(X_{H,z}) = z^2 \gamma_{\star, \eta_\star} < \Gamma_{90}^{\text{Ge}}.$$

Every retained and enriched witness belongs to the common physical cone with apex $W_{H,z}^\star$. Physical correction deletion returns W_H without retuning. The point $z = 0$ is a defined zero-amplitude destructive-interference survivor, not an inference from non-observation or source deletion.

Proof. D2-02 generates the compiled source before response evaluation. [Theorems 6.4, 6.9](#) and [6.11](#) construct one constraint-preserving semantic physical apex, prove its exact D2 transition and physical base restriction, and prove every inherited, pure-process, cross, and common-cone residual. [Theorem 6.6](#) and [corollary 6.7](#) independently prove in-situ equality of the complete frozen oscillation evaluator and seven-channel standing packet, without correction deletion. The accepted-frame condition and all applicable residuals follow from [lemma 6.12](#). $\square$

Corollary 7.2 (Uniform defined-zero lift). *For every framed base realization W_H , $X_{H,0}$ has exact transition block zero and is a defined zero-response survivor at $\eta_\star$, with physical base restriction W_H . Its carrier trace is $P_{H,0}^{\star, \text{red}} = \text{ActRed}(P_{H, -A_H})$: it is the compiled trace when $-A_H \neq 0$, and it reduces to H when $A_H = 0$. No source-null correction block is retained merely to manufacture multiplicity.*

8 Restriction to the fixed oscillation quotient

8.1 Standing restriction

The primary PCR-5C restriction is defined from the *in-situ retained old-target standing core*, not from physical correction deletion. On the valid-reduct subgroupoid, define

$$R_{D2} : \mathcal{N}(\widehat{T}_{D2}^{\text{Ge}}) \longrightarrow \mathcal{N}(T_{\text{osc}}), \quad R_{D2}(W) := \text{Core}_{\text{osc}}\left(\Phi_4^{\text{ret}}(W), \Xi_{\text{osc}}^{\text{ret}}(W)\right).$$

For compiled transparent points, [theorem 6.6](#) gives $R_{D2}(W_{H,C}) \simeq \text{Core}_{\text{osc}}(W_H)$. The physical retraction Res_{osc} and formal reduct U_{D2}^{CMP} are secondary compatibility maps; [theorem 6.4](#) proves that they commute with the carrier trace, while [theorem 6.6](#) proves that they agree with the primary standing restriction on the transparent compiled locus. On arrows, retain every old-target standing witness and continuation before reducing to the fixed base core.

Lemma 8.1 (Representative independence). *If $X \simeq_{D2} Y$ in $\mathcal{N}(\hat{T}_{D2}^{\text{Ge}})$, then $R_{D2}(X) \simeq_{\text{osc}} R_{D2}(Y)$ in $\mathcal{N}(T_{\text{osc}})$. Hence*

$$\rho_{D2} = \pi_0(R_{D2}) : Q(\hat{T}_{D2}^{\text{Ge}}) \longrightarrow Q_{\text{osc}}^{\text{SCIC}}$$

is well defined.

Proof. An enriched SCIC equivalence preserves every retained standing witness and continuation. Restriction therefore preserves equality of the frozen base continuation signature. PCR-5B and SCIC identify equality of that signature with base standing-core equivalence. $\square$

Theorem 8.2 (D2-04: restriction and no merger). *The functor R_{D2} is compatible with the source reduct and B15 framing. Distinct base classes cannot merge:*

$$q_1 \neq q_2 \implies \rho_{D2}^{-1}(q_1) \cap \rho_{D2}^{-1}(q_2) = \emptyset.$$

The four permanent PCR-5A law/new-local deletion classes remain pairwise distinct after D2 extension.

Proof. The Source-Faithful Enriched Realization Theorem gives the natural commuting square between enriched physical objects, carrier traces, physical base restrictions, and formal source reducts. The Retained-Oscillation Transparency and Base-Conservativity Theorem separately proves that, on the compiled physical locus, the in-situ retained evaluator equals the complete base evaluator channel by channel while the correction remains present. Thus the primary standing restriction, physical retraction, and formal reduct agree there. If an enriched component restricted to two distinct base components, an enriched equivalence would identify a frozen base separator, contradicting retained-witness covariance and SCIC uniqueness. The frozen deletion entries remain retained separators, so the four marked classes cannot merge. $\square$

8.2 Choice-dependent section

Choose one B15 frame H_q for each q and one $z_0 \in I_{76,H_q}^{\star,\text{act}}$. By [corollary 6.7](#), $C_{H_q,z_0}^{\star} \in \mathcal{V}_{76}^{\text{tr}}(H_q)$. Then

$$s_{z_0}(q) = [X_{H_q,z_0}]$$

is a set-theoretic right inverse of ρ_{D2} . It depends on the chosen base frame, source-amplitude presentation, exact-amplitude process frame, and z_0 ; no canonical representative-independent enriched identity is claimed.

9 Exact D2 lift fibres

9.1 Connected-class fibre

For $q \in Q_{\text{osc}}^{\text{SCIC}}$, let $\mathcal{N}(T_{\text{osc}})[q]$ be its full connected subgroupoid and define

$$\mathcal{L}_{D2}(q) = \mathcal{N}(\hat{T}_{D2}^{\text{Ge}}) \times_{\mathcal{N}(T_{\text{osc}})}^h \mathcal{N}(T_{\text{osc}})[q].$$

For a frame $n \in \mathcal{N}(T_{\text{osc}})[q]$, the framed fibre satisfies

$$\mathcal{L}_{D2}(q) \simeq (\mathcal{N}(\hat{T}_{D2}^{\text{Ge}}) \times_{\mathcal{N}(T_{\text{osc}})}^h \{n\}) // \text{Aut}_{\mathcal{N}(T_{\text{osc}})}(n).$$

This is the primary groupoid object; cardinality statements are made only after applying π_0 .

9.2 Exact chart datum

For every PCR-5B source skeleton $a = (F, \sigma)$ and for the compiled chart, define the complete chart datum

$$\mathbf{H}_a^{D2} = (\Theta_a^{D2, \text{raw}}, E_a^{D2}, A_a^{D2}, \text{Stat}_a^{D2}, Z_a^{D2}, \Gamma_a^{D2}, p_a^{D2}, \mathcal{M}_a^{D2}, \text{Prov}_a, \Delta_{\text{fr}}^{\text{del}}, \text{Comp}_a^{D2}, \partial \text{Stat}_a^{D2}, \text{Sep}_a^{D2}, \text{Ov}_a^{D2}).$$

The new burden is not discharged by writing a list of residual names. Each family must provide an actual exact source-amplitude map and its target response.

For every family map $A_{a,F}$ and complete response frame $\eta \in \mathcal{E}_{76}^{\text{adm}}$, define

$$K_{a,F,\eta}(\theta) := \mathcal{R}_{76,\eta} A_{a,F}(\theta), \quad \Gamma_{a,F,\eta}(\theta) := \mathfrak{G}_{76,\eta}(K_{a,F,\eta}(\theta), K_{a,F,\eta}(\theta)).$$

The complete phenomenon-specific residual is the explicit fibred map

$$\begin{aligned} \mathcal{E}_{a,F,\eta}^{D2}(\theta, A, K, \Gamma) := & (E_a^{B15}(\theta), E_F^{\text{src}}(\theta), A - A_{a,F}(\theta), E_{\text{amp}}(A), \\ & K - \mathcal{R}_{76,\eta} A, \Gamma - \mathfrak{G}_{76,\eta}(K, K), E_{\text{frame}}(\eta), E_{\text{app}}^F(\theta, \eta), \\ & E_{\text{inst}}(K, \eta), E_{\text{red}}^F(\theta), E_{\text{cone}}^F(\theta, A, \eta), 1 - \text{Acc}_{90,\eta}(K)). \end{aligned} \quad (19)$$

where

$$E_{\text{amp}}(A) = 0 \iff A \in \mathcal{A}_{76}^{\text{phys}}.$$

Here E_F^{src} is the exact fixed PCR-5B family equation system—L1–L5, N1–N5, H1–H5, C1–C5, SO1–SO5, or Y1–Y7—not a new registry label. Applicability is fixed before evaluation. The pass object is the guarded zero locus of every equation-valued coordinate together with defined instrument response and the accepted complete-frame verdict; the ordered first decisive non-pass coordinate gives the exact typed complement. In particular, a source-coherent exact amplitude outside the contraction locus is a physical-amplitude failure, not N/A and not a candidate-generation deletion.

9.3 Explicit six-family exact-amplitude maps

For each finite skeleton, the following formulas are finite sums or contractions in the source-defined parameter object. They are therefore well defined symbolically even when the parameter object is functional or categorical.

LAW. For the PCR-5B LAW path set $\Pi_{a,\text{LAW}}^\lambda$,

$$A_{a,\text{LAW}}(\theta) = \bigoplus_{\lambda} \sum_{\pi \in \Pi_{a,\text{LAW}}^\lambda} A_\pi^f L_\pi(\theta_L, \text{Mem}, \text{Match}) A_\pi^i.$$

The operators L_π are the actual ActLaw/process morphisms of L1–L5, including retained memory, order, threshold, and matching data.

LOCAL. For each oriented local source path $\pi = (e_1, v_1, \dots, v_{r-1}, e_r)$, let

$$G_v(p; \theta) = (\not{p} - M_v(\theta) - \Sigma_v^{\text{ind}}(p; \theta) + i0)^{-1}.$$

Then

$$A_{a,\text{LOCAL}}(\theta) = \bigoplus_{\lambda} \sum_{\pi} A_\pi^f V_{e_r}(\theta) G_{v_{r-1}} \cdots G_{v_1} V_{e_1}(\theta) A_\pi^i + A_{a,\text{LOCAL}}^{\text{contact/ind}}(\theta).$$

The last term is the finite sum of source-declared contact, radiative, and black-box-induced contributions. It is part of the same local source point and cannot be discarded in the compiled chart.

HID. Choose any Stinespring chart $(V_\theta, \Omega_\theta)$ for the active hidden process and an orthonormal basis $\{r\}$ of its minimal source-generated hidden support. The purified exact amplitude is

$$A_{a,\text{HID}}(\theta) = \bigoplus_{\lambda, r} A_\lambda^f P_{\text{vis}} \langle r | V_\theta | \Omega_\theta \rangle P_{\text{in}} A_\lambda^i.$$

Changing the Stinespring chart acts unitarily on the direct-sum label and is a lawful groupoid arrow. Memory and population maps are composed in source order before the visible projection.

COMP. For every constituent/derived path π , let $B_\pi(\theta)$ be the source-coherent binding/evolution map and D_π^i, D_π^f the declared derivation maps. Then

$$A_{a,\text{COMP}}(\theta) = \bigoplus_{\lambda} \sum_{\pi \in \Pi_{a,\text{COMP}}^\lambda} A_\pi^f D_\pi^f(\theta) B_\pi(\theta) D_\pi^i(\theta) A_\pi^i.$$

Every constituent retained by C1–C5 appears in at least one contraction or in the exact deletion/failure trace.

SOL. For a solved profile family $\phi_\theta(\xi)$ on source-defined collective-coordinate space $\mathcal{M}_\phi$,

$$A_{a,\text{SOL}}(\theta) = \bigoplus_{\lambda} \sum_{\pi} A_\pi^f \int_{\mathcal{M}_\phi} \langle \phi_f(\xi) | \mathcal{O}_{76,\pi}[\phi_\theta, \xi] | \phi_i(\xi) \rangle d\mu_\theta(\xi) A_\pi^i.$$

The source equation, finite energy/support, stability, persistence, endability, attachment, and continuation conditions are SO1–SO5.

HYB. Let $T_e(\theta)$ be the declared multilinear map of each active higher edge. The decorated-hybrid exact amplitude is

$$A_{a,\text{HYB}}(\theta) = \bigoplus_{\lambda \in \Lambda_a^{\text{inc}}} \sum_{\pi \in \Pi_{a,\text{HYB}}^\lambda} \text{Contr}_\pi(\{A_i(\theta)\}_{i \in I}, \{T_e(\theta)\}_{e \in E_{\text{high}}}, \text{Glue}(\theta)).$$

Coherent paths are summed in one sector; source-declared incoherent alternatives occupy orthogonal purified sectors. All Inflow, Fuse, CoCarry, Project, boundary/edge, and higher-arity relations required by Y1–Y7 remain in the contraction.

Theorem 9.1 (Executed family response-map theorem). *For every PCR-5B skeleton a , the formulas above and [equation \(19\)](#) give a total, source-semantic, lawful-groupoid-covariant D2 map. Every semantic intrinsic D2 response in the declared comparison class is represented by exactly one family/skeleton amplitude up to source overlap, and every generated exact amplitude has a coherent physical source interpretation. The status partition is exact and no active failure or undefined response is reclassified as N/A.*

Proof. The source-amplitude normal-form theorem gives coverage and soundness by structural induction. The displayed formulas are its family specializations and use the actual PCR-5B source equations. For each complete frame, the fibred map $\mathcal{R}_{76,\eta}$ and Hermitian form $\mathfrak{G}_{76,\eta}$ compute the kernel and rate by composition rather than by a lookup label; [theorem 4.1](#) proves lawful frame covariance. PCR-4/B15 supplies the base residual, exact applicability, source reduct, and common-cone data. The ordered total compiler then partitions every raw point into pass, certified N/A, failure, undefined, response out of scope, source update, or target update. Covariance of every exact-amplitude contraction under the family groupoid proves descent. $\square$

9.4 Compiled chart

The source-complete compiled chart is evaluated on

$$\Theta_{a,\text{CMP}}^{D2,\text{raw}} = \Theta_a^{\text{raw}} \times \mathcal{V}_{76}^{\text{src}} \times \mathcal{E}_{76}^{\text{adm}}.$$

For $H_a(\theta)$ and arbitrary $C \in \mathcal{V}_{76}^{\text{src}}$, its exact complete amplitude, event kernel, and rate are

$$A_{a,\text{CMP}}(\theta, C) = A_{a,F}(\theta) + C,$$

$$K_{a,\text{CMP},\eta}(\theta, C) = \mathcal{R}_{76,\eta}(A_{a,F}(\theta) + C), \quad \Gamma_{a,\text{CMP},\eta}(\theta, C) = \mathfrak{G}_{76,\eta}(K_{a,\text{CMP},\eta}(\theta, C), K_{a,\text{CMP},\eta}(\theta, C)).$$

The exact prequotient pass object is

$$\tilde{Z}_{a,\text{CMP}}^{D2} = \left\{ (\theta, C, \eta) \left| \begin{array}{l} \theta \in Z_a, \quad C \in \mathcal{V}_{76}^{\text{src}}, \quad \eta \in \mathcal{E}_{76}^{\text{adm}}, \\ A_{a,F}(\theta) + C \in \mathcal{A}_{76}^{\text{phys}}, \\ W_{H_a(\theta),C} \in \text{PhysMod}_{4,D2}^{\text{sem}}, \\ C \in \mathcal{V}_{76}^{\text{tr}}(H_a(\theta)), \\ \mathcal{R}_{76,\eta}(A_{a,F}(\theta) + C) \text{ is defined,} \\ \text{Acc}_{90,\eta}(K_{a,\text{CMP},\eta}(\theta, C)) = 1 \end{array} \right. \right\}.$$

Its lawful quotient uses the framed base groupoid, process arrows, source-module arrows, and $\Gamma_{76}^{\text{resp}}$. Only after this complete chart exists does the response theorem mark

$$C_{a,\theta,z}^* = -A_{a,F}(\theta) + zA_{76}^*, \quad z \in [0, 1].$$

The marked accepted subchart

$$\{(\theta, C_{a,\theta,z}^*, \eta_*) : \theta \in Z_a, \quad z \in I_{76,\eta_*}^{\text{acc}}\} \subset \tilde{Z}_{a,\text{CMP}}^{D2}$$

is used for the universal continuum injection. The exact transparent-chart complement consists of base failure, source-extension failure, nonphysical or ill-typed complete amplitude $A_{a,F}(\theta) + C$, semantic-amalgamation failure, frame failure, undefined/ROOS response, or failure of the complete-frame acceptance predicate. A point with an activated old-target incidence is not counted as a physical failure merely for being nontransparent: $\text{Route}_{\text{osc}}$ sends its in-situ retained packet to the corresponding base core when one exists, otherwise to target update. No response-visible direction defines the raw domain.

Theorem 9.2 (Compiled-chart classification). *The preceding fibred datum is exact for every B15 skeleton. It retains every intrinsic D2 contribution in the one-point amplitude, every solved transparent point is the source-faithful physical object $W_{H_a(\theta),C}$ with $C \in \mathcal{V}_{76}^{\text{tr}}(H_a(\theta))$, its response descends under $\Gamma_{76}^{\text{resp}}$, and its solved object contains precisely all accepted transparent compiled survivors from the complete source-defined C -domain over that skeleton. Nontransparent source presentations are retained in the global D2 universe through their intrinsic/HYB routing rather than being converted into failed points of this chart. The marked affine family is a solved subchart used only for nonemptiness and the continuum lower bound. Its lawful reduct is the projection to Z_a .*

Proof. D2-02 and [theorems 5.1, 6.1, 6.4 and 6.11](#) give source coverage, source-faithful physical realization, complete interference, physical restriction, and exact reduct. D2-01 and [theorem 4.1](#) give total framewise status and lawful descent. The general compiled exact amplitude is $A_{a,F}(\theta) + C$; the marked cancellation identity is applied only on the downstream-selected witness subchart. Hence every raw triple (θ, C, η) lies in exactly one survivor or typed complement cell, independently of whether it belongs to the marked line. $\square$

9.5 LAW chart

The LAW raw domain is the PCR-5B L1–L5 source domain augmented by $(a_{76}, \eta, A, K, \Gamma)$. Its exact residual is [equation \(19\)](#) with $A_{a, \text{LAW}}$ above. Applicability routes every defined mechanism-neutral law response through EXP_RATE and activates LME or SMEFT_D9 only on their proved domains. The lawful groupoid is the PCR-5B LAW groupoid acting on $A_{a, \text{LAW}}$, the nuclear/detector chart, and the framed D2 core.

Theorem 9.3 (Exact LAW D2 classification). *The guarded zero locus of $\mathcal{E}_{a, \text{LAW}}^{D2}$ is exactly the LAW D2 survivor object. Its quotient retains every standing-visible kernel, rate, memory, matching, continuation, deletion, and failure difference; its ordered complement is exact.*

Proof. L1–L5 generate the complete LAW source trace. The explicit finite contraction $A_{a, \text{LAW}}$ computes every D2 path of that trace, and [theorem 9.1](#) proves coverage, soundness, total status, and lawful descent. $\square$

9.6 LOCAL chart

The LOCAL raw domain is the PCR-5B N1–N5 source domain with representation, charge, mass, vertex, threshold, vacuum, matching, induced-term, and D2 response variables. The map $A_{a, \text{LOCAL}}$ is the finite path sum of actual vertices, full propagators, contact terms, and every source-required induced contribution. Applicability activates the local-QFT, LME, or SMEFT specialization only where its hypotheses hold.

Theorem 9.4 (Exact LOCAL D2 classification). *The guarded zero locus of $\mathcal{E}_{a, \text{LOCAL}}^{D2}$ is exactly the local one-point survivor locus. The quotient retains active rank, representation, thresholds, oriented couplings/matching, induced continuation, kernel/rate, deletion, and failure data; the complement is the exact union of source, anomaly, vacuum, threshold, induced, matching, response, and update cells.*

Proof. N1–N5 and the B15 residual fix the source point and shared physical ledger. The explicit resolvent/vertex/contact formula computes its complete D2 amplitude, including black-box and radiative terms. [Theorem 9.1](#) then gives semantic exactness and descent. $\square$

9.7 HID chart

The HID raw domain is the PCR-5B H1–H5 minimal active hidden-support domain with dilation, projection, memory, population, response, reduct, and status data. The Stinespring/Kraus amplitude above is evaluated on that same minimal support.

Theorem 9.5 (Exact HID D2 classification). *The guarded zero locus of $\mathcal{E}_{a, \text{HID}}^{D2}$ is exactly the coherent active hidden-support survivor locus. Hidden spectators, malformed dilations, undefined projections, response-out-of-scope outputs, reduct failures, and simultaneous failures form the exact complement. Hidden support, projection, memory, kernel/rate, deletion, and failure frontiers are retained separators.*

Proof. H1–H5 construct the minimal source support. The purified Kraus amplitude is invariant under Stinespring change and computes the complete visible D2 response. Exactness follows from [theorem 9.1](#). $\square$

9.8 COMP chart

The COMP raw domain is the PCR-5B C1–C5 constituent/derived domain with binding, derivation, support, matching, D2 response, reduct, and status data. The exact amplitude is the explicit finite sum $D_f B D_i$ over all source-active constituent paths.

Theorem 9.6 (Exact COMP D2 classification). *The guarded zero locus of $\mathcal{E}_{a,\text{COMP}}^{D2}$ is exactly the constituent-retaining, coherently bound, derivation-complete, support/matching-compatible D2 survivor locus. Constituent support, binding, derivation, matching, kernel/rate, deletion, and failure data classify its standing moduli and complement.*

Proof. C1–C5 make the constituent and derivation maps source data on one point. The displayed finite contraction computes every target-bearing derived path. Apply [theorem 9.1](#). $\square$

9.9 SOL chart

The SOL raw domain is the PCR-5B SO1–SO5 profile domain with source equation, finite support/energy, topology, persistence, stability, endability, attachment, continuation, D2 matrix element, and status data.

Theorem 9.7 (Exact SOL D2 classification). *The guarded zero locus of $\mathcal{E}_{a,\text{SOL}}^{D2}$ is exactly the source-coherent finite-profile D2 survivor locus. Profile, topology, persistence, endability, attachment, continuation, kernel/rate, deletion, and failure data form the exact moduli and complement.*

Proof. SO1–SO5 construct the profile and its source-defined measure/continuation data. The displayed matrix-element integral computes its complete D2 amplitude. Apply [theorem 9.1](#). $\square$

9.10 HYB chart

The HYB raw domain is the PCR-5B Y1–Y7 decorated hypergraph domain, including every source-active Inflow, Fuse, CoCarry, Project, boundary/edge, and finite higher relation. The response amplitude is the direct decorated-hyperpath contraction above; no pairwise substitute is used for a primitive higher edge.

Theorem 9.8 (Exact HYB D2 classification). *The guarded zero locus of $\mathcal{E}_{a,\text{HYB}}^{D2}$ is exactly the source-coherent decorated-hybrid D2 survivor locus. Its moduli retain every standing-visible constituent, higher relation, interference or purified-incoherent sector, gluing, reduct, cone, deletion, status, and isotropy difference. Every non-survivor has an exact constituent, higher-relation, gluing, response, cone, or update status.*

Proof. Y1–Y7 generate and retain the full decorated source object. Structural recursion evaluates every active constituent and higher edge on one point. The source-amplitude normal form and [theorem 9.1](#) prove coverage, soundness, and lawful descent. $\square$

9.11 Uniform accepted interval and global fibre

For each framed B15 physical realization W_H and $z \in I_{76,H}^{\star,\text{act}}$, D2-03 supplies the constraint-preserving source-faithful survivor $X_{H,z} = (W_{H,z}^{\star}, \eta_{\star})$. If $z_1 \neq z_2$, then

$$\Gamma_{1/2}^{0\nu}(H, z_1, \eta_{\star}) = z_1^2 \gamma_{\star, \eta_{\star}} \neq z_2^2 \gamma_{\star, \eta_{\star}} = \Gamma_{1/2}^{0\nu}(H, z_2, \eta_{\star}),$$

so the two points are standing distinct at the same complete response frame. The active locus $I_{76,H}^{\star,\text{act}}$ is obtained from a nondegenerate real interval by deleting at most one point, so it has cardinality $\mathfrak{c}$.

Theorem 9.9 (D2-05: executed fibre exhaustion and coarse type). *For every $q \in Q_{\text{osc}}^{\text{SCIC}}$:*

1. *the compiled, LAW, LOCAL, HID, COMP, SOL, and HYB solved groupoids cover every source-lawful D2 survivor framed over $\mathcal{N}(T_{\text{osc}})[q]$;*

2. every object in those solved groupoids is a one-point D2 survivor and every complement is exactly typed;
3. duplicate presentations are identified only by the recomputed pairwise and higher complete-core systems of D2-10;
4. for every chosen framed H over q , $\mathcal{L}_{D2}(q) \neq \emptyset$ and $z \mapsto [X_{H,z}]$ embeds $I_{76,H}^{\star, \text{act}}$ into $\pi_0 \mathcal{L}_{D2}(q)$;
5. therefore $|\pi_0 \mathcal{L}_{D2}(q)| \geq \mathfrak{c}$;
6. the exact coarse type is **ParameterizedMixed**: continuous response moduli coexist, where present, with discrete, functional, categorical, interference, provenance, and isotropy structure;
7. no comparison-class-independent upper cardinal follows.

Proof. D2-02 and [theorems 5.1](#) and [9.1](#) give semantic source coverage and explicit source-to-response execution for every family. The seven chart theorems give the exact solved objects and complements. D2-10 supplies the complete duplicate-presentation relation and higher coherence. The complete compiled chart supplies source coverage over all $C \in \mathcal{V}_{76}^{\text{src}}$. Inside that already-complete chart, the marked affine subfamily proves nonemptiness over every PCR-5B/B15 base frame at the response witness frame $\eta_\star$, and the physical rate injects the continuum active locus $I_{76,H}^{\star, \text{act}}$. Arbitrary admitted symbolic parameter objects remain possible in the intrinsic charts, so the interval is a lower bound rather than an upper-cardinality theorem. $\square$

Corollary 9.10 (Global decomposition). *There is an exact coarse decomposition*

$$Q(\hat{T}_{D2}^{\text{Ge}}) \cong \bigsqcup_{q \in Q_{\text{osc}}^{\text{SCIC}}} \pi_0 \mathcal{L}_{D2}(q),$$

and the enriched groupoid is reconstructed by the Grothendieck construction of the executed D2 lift pseudofunctor over $\mathcal{N}(T_{\text{osc}})$.

10 Nuclear-response and parameter descent

10.1 Six parameter roles

Every D2 coordinate has one primary role fixed before carrier counting:

1. **source constitutive**: changes the physical source trace;
2. **empirical nuisance**: belongs to the detector/statistical model inside one complete frame;
3. **nuclear response**: belongs to η_{nuc} and maps exact source amplitude to the nuclear transition object;
4. **lawful presentation**: changes only the representative inside a $\Gamma_{76}^{\text{resp}}$ or source-groupoid orbit;
5. **standing separator**: changes a descended target-authorized physical consequence;
6. **update**: changes source theory, isotope, event topology, data release, response primitive, or target scope.

One numerical symbol may appear in several formulas but acquires carrier identity only through a standing-visible consequence.

10.2 Complete response descent

The response object is the fibred section

$$\eta \longmapsto (\mathcal{R}_{76,\eta}, \mathfrak{G}_{76,\eta}, \mathcal{I}_{76,\eta}, \mathcal{L}_{76,\eta}, \text{Acc}_{90,\eta})$$

over $\mathcal{E}_{76}^{\text{adm}}$, modulo $\Gamma_{76}^{\text{resp}}$. For an exact source amplitude A , its descended response record is the groupoid-valued section

$$\overline{\text{Resp}}_{76}(A) = [\eta \longmapsto (\mathcal{R}_{76,\eta}A, \mathfrak{G}_{76,\eta}(\mathcal{R}_{76,\eta}A, \mathcal{R}_{76,\eta}A), \mathcal{I}_{76,\eta}(\mathcal{R}_{76,\eta}A), \mathcal{L}_{76,\eta}, \text{Acc}_{90,\eta}(\mathcal{R}_{76,\eta}A))] / \Gamma_{76}^{\text{resp}}.$$

The candidate survives exactly by the pre-fixed existential complete-frame condition (6). The quotient does not collapse physically inequivalent nuclear or detector frames, and existence may not be proved by splicing coordinates from different frames.

Lemma 10.1 (Nuisance covariance). *Lawful response arrows preserve applicability, physical rate, likelihood ordering, acceptance, and D2 standing status. A transformation that changes any such output is not a nuisance arrow.*

Proof. This is [theorem 4.1](#) specialized to the D2 parameter-role partition. $\square$

10.3 Nuclear matrix elements do not select carriers

Nuclear uncertainty may deform the accepted source-parameter locus and obstruct a sharp inverse microscopic inference. It is not itself multiplicity of neutrino carriers. A nuclear-method label or sampled NME value refines standing identity only when the complete descended response or licensed continuation changes.

Theorem 10.2 (D2-06: parameter and nuclear-response descent). *At the frozen D2 target:*

1. every empirical, nuclear, EFT, source, chart, nuisance, and update coordinate has one of the six roles above;
2. lawful response presentations descend under $\Gamma_{76}^{\text{resp}}$ while physically inequivalent complete frames remain distinct response possibilities;
3. an NME, phase-space factor, hadronic LEC, detector nuisance, likelihood coordinate, or many-body label refines carrier identity only through a changed descended physical response or continuation;
4. LME and SMEFT parameters exist only on their certified loci;
5. uncertainty in the inverse rate-to-microscopic map cannot eliminate a source class whose mechanism-neutral response survives in one lawful complete frame.

Proof. D2-01 and D2-02 give exhaustive target/source roles. Response covariance gives quotient descent, while (6) fixes the candidate-level quantifier. Any nonfactorizing variation changes source trace, standing response, or scope. Specialized formulas are partial charts with explicit hypotheses, and ambiguity of a reverse inference is not forward-response failure. $\square$

Corollary 10.3 (No NME preference theorem). *A preferred NME calculation, narrower uncertainty interval, or familiar many-body method cannot select a PCR-5B base class by itself.*

11 Standing-amplitude descent and enriched SCIC typing

Let $\mathcal{W}_{D2}^A$ be the independently admitted D2 standing-witness family whose values may depend on an exact transition-amplitude presentation, and let $\mathcal{U}_{D2}$ be the licensed continuation category. The comparison is fibred over the retained base core:

$$\tilde{\mathcal{A}}_{76}^{\text{phys}} = \{(n, X, A_X) : n \in \mathcal{N}(T_{\text{osc}}), X \text{ is a framed D2 survivor over } n, A_X \in \mathcal{A}_{76}^{\text{phys}}\}.$$

Define

$$\Sigma_{D2}^A(X) = (w(uX))_{w \in \mathcal{W}_{D2}^A, u \in \mathcal{U}_{D2}}.$$

Definition 11.1 (Standing-amplitude equivalence). *Two fibred exact-amplitude presentations (n, X, A_X) and (n', Y, A_Y) are equivalent when a lawful retained-base isomorphism $\alpha : n \rightarrow n'$ transports every coordinate of $\Sigma_{D2}^A(X)$ to the corresponding coordinate of $\Sigma_{D2}^A(Y)$. The quotient is*

$$\mathcal{A}_{76}^{\text{stand}} := \tilde{\mathcal{A}}_{76}^{\text{phys}} / \sim_{D2}^{A, \text{St}}.$$

The notation $[A_X]_{D2}^{\text{St}}$ means the class of the complete fibred presentation; raw amplitude representatives remain framed provenance and have no independent terminal identity authority.

Theorem 11.2 (Standing-amplitude SCIC descent). *The relation $\sim_{D2}^{A, \text{St}}$ is the unique admissible same-scope amplitude identity relation compatible with the frozen enriched witness and continuation grammar. Raw amplitude equality is sufficient but not necessary. A difference invisible to every admitted witness after every licensed continuation collapses; a difference in response, attachment, matching, continuation, deletion, composition, status, failure, or higher incidence remains standing. No provenance label, representative choice, or auxiliary amplitude norm may refine this quotient.*

Proof. Apply SCIC to the amplitude-sensitive portion of the activated D2 witness algebra. Witness soundness forbids merging a detected difference, while witness completeness and no-second-classifier closure forbid retaining a difference with no licensed consequence. Continuation stability follows from closure of $\mathcal{U}_{D2}$. $\square$

The complete enriched continuation signature is

$$\Sigma_{D2}^{\text{enr}}(X) = \left(\Xi_{\text{osc}}^{\text{ret}}(X), [A_X]_{D2}^{\text{St}}, \overline{\text{Resp}}_{76}(A_X), \text{Att}_X, \text{Match}_X, \text{Cont}_X, \text{Comp}_X, \partial \text{Stat}_X, \text{High}_X \right)_{u \in \mathcal{U}_{D2}}.$$

12 Enriched source-composition structure

12.1 Two exact deletion problems

Let X be a D2 survivor with finite active source-block set $B(X)$. For every constitutively well-typed subset $S \subseteq B(X)$, let $\text{Del}_{B(X) \setminus S} X$ delete the complement with all retained source parameters fixed. No coupling is retuned, no deleted block is replaced, no new incidence is generated, and no response chart or correction amplitude is recomputed.

Define the identity-preserving predicate

$$N_{D2, X}^{\text{id}}(S) = \begin{cases} 1, & \Sigma_{D2}^{\text{enr}}(\text{Del}_{B(X) \setminus S} X) = \Sigma_{D2}^{\text{enr}}(X), \\ 0, & \text{the deletion is defined but changes or fails the enriched signature,} \\ \perp, & \text{the retained source is constitutively ill typed,} \end{cases}$$

and the coarser target-survival predicate

$$N_{D2,X}^{\text{surv}}(S) = \begin{cases} 1, & \text{Stat}_{D2}(\text{Del}_{B(X)\setminus S} X) = \text{Pass}, \\ 0, & \text{the defined deletion is not a D2 survivor}, \\ \perp, & \text{the retained source is ill typed.} \end{cases}$$

The first determines enriched carrier identity and S/R/P/Y composition. The second records only whether the present null-search target still survives; it has no authority to refine SCIC.

Let

$$\mathcal{M}_{\text{id}}(X) = \min_{\subseteq} \{S : N_{D2,X}^{\text{id}}(S) = 1\}, \quad \mathcal{M}_{\text{surv}}(X) = \min_{\subseteq} \{S : N_{D2,X}^{\text{surv}}(S) = 1\}.$$

Since X is a survivor, $B(X)$ belongs to both domains, so both minimal-set families are nonempty.

12.2 Disjoint S/R/P/Y rule

The exact chartwise code is

$$\text{Code}_{D2}(X) = \begin{cases} \text{S}, & \mathcal{M}_{\text{id}}(X) = \{S_0\}, |S_0| = 1, \\ \text{R}, & |\mathcal{M}_{\text{id}}(X)| \geq 2 \text{ and some } S \in \mathcal{M}_{\text{id}}(X) \text{ has } |S| = 1, \\ \text{Y}, & \mathcal{M}_{\text{id}}(X) = \{B(X)\}, |B(X)| > 1, \\ \text{P}, & \text{all remaining cases; every minimal route contains multiple blocks.} \end{cases}$$

The cases are disjoint and exhaustive. The **P** branch includes several incomparable multi-block minima and a unique proper multi-block minimum with additional identity-redundant active blocks.

12.3 Compiled-chart deletion at fixed correction amplitude

For a compiled physical survivor $W_{H,C}$, its carrier trace is $P_{H,C}^{\text{red}}$. Its active block set is

$$B(P_{H,C}^{\text{red}}) = \begin{cases} B(H) \sqcup \{c_{76}\}, & C \neq 0, \\ B(H), & C = 0. \end{cases}$$

The full physical correction deletion is fixed by

$$\text{Del}_{c_{76}}^{\text{phys}}(W_{H,C}) \simeq W_H, \quad \chi_{\text{car}}(\text{Del}_{c_{76}}^{\text{phys}}(W_{H,C})) \simeq U_{D2}^{\text{CMP}}(P_{H,C}^{\text{red}}) = H.$$

For the finer source-subset calculation, where c_{76} is the correction-amplitude block, put

$$A_H^{\text{full}} = \mathcal{A}_{76}^{\text{src}}(H), \quad A_H(S) = \mathcal{A}_{76}^{\text{src}}(\text{Del}_{B(H)\setminus S} H).$$

No-repair deletion keeps the original correction parameter fixed:

$$A_{\text{corr}}^{\text{fixed}}(H, C) = C.$$

On the marked witness, $C = -A_H^{\text{full}} + zA_{76}^{\star}$. For $S \subseteq B(H)$ and $\epsilon \in \{0, 1\}$, let

$$X_{S,\epsilon} = \text{Del}_{B(H)\setminus S} H \boxplus \epsilon c_{76}.$$

Its exact complete amplitude and framewise kernel are

$$A_{S,\epsilon} = A_H(S) + \epsilon C, \quad K_{S,\epsilon,\eta} = \mathcal{R}_{76,\eta}(A_{S,\epsilon}).$$

Physicality requires $A_{S,\epsilon} \in \mathcal{A}_{76}^{\text{phys}}$, and target survival is evaluated by the same complete-frame predicate as every other D2 point. Thus the undeleted amplitude is $A_H^{\text{full}} + C$; deleting the correction gives $A_H(S)$; deleting base blocks while retaining the marked correction leaves the no-repair mismatch

$$A_H(S) - A_H^{\text{full}} + zA_{76}^*.$$

No correction is recomputed to cancel the reduced source.

Define

$$\begin{aligned} \mathfrak{D}_{H,C}^{\text{id}} &= \{(S, \epsilon) : X_{S,\epsilon} \text{ is typed and } \Sigma_{D2}^{\text{enr}}(X_{S,\epsilon}) = \Sigma_{D2}^{\text{enr}}(P_{H,C}^{\text{red}})\}, \\ \mathfrak{D}_{H,C}^{\text{surv}} &= \{(S, \epsilon) : X_{S,\epsilon} \text{ is typed and } \text{Stat}_{D2}(X_{S,\epsilon}) = \text{Pass}\}. \end{aligned}$$

The correction block is identity-essential or survival-essential exactly when every corresponding inclusion-minimal subset retains $\epsilon = 1$.

12.4 Intrinsic chart rules

The exact block sets and deletion tests are determined by the solved amplitude maps:

Band	Active blocks	Exact deletion test
LAW	law/process nodes, memory/continuation cells, matching and D2 attachment	Recompute L1–L5, the explicit $A_{a,\text{LAW}}$ contraction, nuclear/detector response, matching, continuation, complete signature, and status without retuning.
LOCAL	local carriers, vertices, vacuum/threshold blocks, induced-term and matching cells	Recompute N1–N5, every propagator/vertex/-contact/induced term in $A_{a,\text{LOCAL}}$, representation/anomaly, vacuum, threshold, matching, response, and cone.
HID	visible owner, hidden support, dilation, projection, memory/population and attachment blocks	Recompute H1–H5, the purified Stine-spring/Kraus amplitude, minimal support, projection, memory, reduct, response, and status.
COMP	constituents, binding, derivation, support, matching and attachment blocks	Recompute C1–C5 and every $D_f B D_i$ contraction, constituent retention, matching, response, reduct, and failure frontier.
SOL	profile, topology, persistence/stability, endability, attachment and continuation blocks	Recompute SO1–SO5 and the profile matrix-element integral, stability/endability, response, continuation, and common cone.
HYB	all constituent blocks and every active pairwise/higher incidence, glue, interference and cone block	Recompute Y1–Y7, every decorated-hyperpath contraction and purified-incoherent sector, exact reduct, cone, response, and status.

Each row is evaluated over the full solved object Z_a^{D2} ; the rule is symbolic and exact at the inherited parameter scope, not a finite sample.

Theorem 12.1 (D2-07: exact enriched source-composition theorem). *For every compiled or intrinsic D2 survivor:*

1. $N_{D2,X}^{\text{id}}$ and $N_{D2,X}^{\text{surv}}$ are total on constitutively typed subsets, fail closed on ill-typed subsets, and invariant under lawful source presentation;

2. the explicit family amplitude maps evaluate every source-block subset and therefore give exact $\mathcal{M}_{\text{id}}(X)$ and $\mathcal{M}_{\text{surv}}(X)$;
3. the disjoint rule above assigns exactly one S/R/P/Y code to every survivor;
4. family names, field counts, naturalness, and base composition labels do not determine the enriched code;
5. physical correction deletion of $W_{H,C}$ returns W_H , and the source-level deletion/reduct of $P_{H,C}^{\text{red}}$ is its carrier-trace shadow; a source-null $C = 0$ block is absent from the active deletion problem; deleting the correction block in the finer subset calculation returns the intrinsic amplitude $A_H(S)$, while retaining it after a base deletion keeps the original correction amplitude fixed and exposes the exact no-repair mismatch;
6. a defined zero-response survivor exists at $z = 0$; its activity-reduced trace is compiled when $-A_H \neq 0$ and is exactly H when $A_H = 0$, so no source-null correction is counted as an active block;
7. no named LNV mechanism is universally forced, but a particular survivor may have a compiled, intrinsic, or mixed block that is identity essential or target-survival essential according to the exact deletion sets.

Proof. Every activity-reduced corrected source presentation has finitely many active blocks and finite higher incidence at the declared occupancy-bearing scope; the block c_{76} occurs exactly when $C \neq 0$. For compiled survivors, [theorem 6.4](#) supplies the corresponding physical object and proves that correction deletion and source reduct commute. Hence every typed subset is covered. PCR-3 source typing decides constitutive definedness, the explicit family amplitude map recomputes the physical response, amplitude descent forms $[A_X]_{D2}^{\text{St}}$, PCR-4/D2 total status evaluates target survival, and the complete enriched SCIC signature decides identity preservation. Lawful source isomorphisms transport subsets, raw amplitudes, standing-amplitude classes, kernels, residuals, signatures, and status. The minimal-set construction therefore executes the exact symbolic classification. The four code branches are disjoint by cardinality and inclusion form and exhaustive because the minimal family is nonempty. $\square$

Remark 12.2 (Future positive target). *A positive future D2 record could remove the zero-response locus and change $\mathcal{M}_{\text{surv}}$. It would be a target update and a new lift calculation, not a retroactive change to the present quotient.*

13 Base image and selection strength

Theorem 13.1 (D2-08: exact base-selection theorem). *The D2 restriction map is surjective:*

$$\boxed{\text{im } \rho_{D2} = Q_{\text{osc}}^{\text{SCIC}}}.$$

Therefore the amplitude ^{76}Ge null-search target eliminates no PCR-5B oscillation class, selects no proper base subset, and does not achieve base singleton selection.

Proof. Let $q \in Q_{\text{osc}}^{\text{SCIC}}$, choose a framed B15 source H_q over it, and choose a semantic physical representative W_{H_q} . Since $I_{76,H_q}^{\star,\text{act}}$ has cardinality $\mathfrak{c}$, choose $z \in I_{76,H_q}^{\star,\text{act}}$. By [corollary 6.7](#), the correction lies in $\mathcal{V}_{76}^{\text{tr}}(H_q)$; D2-03 gives the source-faithful survivor $X_{H_q,z} = (W_{H_q,z}^{\star}, \eta_{\star})$ with $\Xi_{\text{osc}}^{\text{ret}} = \Xi_{\text{osc}}(W_{H_q})$, and D2-04 therefore gives $\rho_{D2}([X_{H_q,z}]) = q$. Hence every base class lies in the image. No merger prevents the construction from coarsening the base quotient. $\square$

The theorem concerns the complete lift fibre, not every individual provenance presentation. Intrinsic LME, EFT, mediator, hidden, composite, solitonic, or hybrid presentations may fail or be undefined; base elimination would require every lawful lift over every framed presentation to fail. The physically realized affine-correction family defeats that condition over every base class.

Corollary 13.2 (No Majorana selection). *The present target does not select light-Majorana exchange, establish that the oscillation carrier is Majorana, or exclude Dirac-protected, exact-zero, or non-LME base-compatible extensions. A conditional $m_{\beta\beta}$ limit remains a response specialization.*

At this target,

$$|\text{im } \rho_{D2}| = |Q_{\text{osc}}^{\text{SCIC}}| \geq \mathfrak{c}, \quad |Q(\hat{T}_{D2}^{\text{Ge}})| \geq \mathfrak{c}.$$

Full enriched singleton selection is impossible because each base fibre already contains continuum many components.

14 Split classes and the D2 standing coordinate

14.1 Relative enriched witness algebra

Let W_{D2}^{rel} be the quotient-factorizing witness algebra added by D2 after removing fixed base witnesses. It includes the physical exact-amplitude/transition-kernel class, inverse-half-life/rate, event-spectrum/partition response, likelihood continuation, specialized-chart applicability, D2 failure status, source-to-nuclear attachment, and D2 deletion consequences. For $y_1, y_2 \in \mathcal{L}_{D2}(q)$, define

$$\text{Sep}_{D2}^{\text{rel}}(y_1, y_2) = \{(w, u) : w(u(y_1)) \neq w(u(y_2))\}.$$

When the enriched witness grammar is not finite or well ordered, the theorem exports the exact nonempty separator family rather than claiming a unique minimal separator.

Theorem 14.1 (Continuum split of every base class). *For every q , every PCR-5B/B15 source frame H over q , and distinct $z_1, z_2 \in I_{76,H}^{\star, \text{act}}$,*

$$[X_{H,z_1}] \neq [X_{H,z_2}] \quad \text{in } \pi_0 \mathcal{L}_{D2}(q).$$

The physical inverse-half-life witness lies in the relative separator family. Thus every base class splits into at least continuum many D2 standing classes.

Proof. Both points have the same in-situ retained base core by [theorem 6.6](#) and [corollary 6.7](#) and D2-04. At the same pre-frozen complete frame $\eta_\star$, their rates $z_1^2 \gamma_{\star, \eta_\star}$ and $z_2^2 \gamma_{\star, \eta_\star}$ differ. By [theorem 4.1](#), this witness is invariant under lawful source and response redescription. Their complete enriched continuation signatures are therefore unequal. The active locus $I_{76,H}^{\star, \text{act}}$ has cardinality $\mathfrak{c}$. $\square$

Definition 14.2 (Fibrewise coordinate independence). *A D2 coordinate is independent of the frozen oscillation coordinate when, over every base class, a standing-nonconstant D2 family exists on which the complete retained oscillation packet is constant. This is weaker than a global product decomposition and does not exclude joint source relations elsewhere.*

Theorem 14.3 (D2-09: split-class and standing-coordinate theorem). *At the frozen target:*

1. *every base class splits;*
2. *every distinct enriched pair has a nonempty exact D2-relative separator family;*

3. the physical rate universally separates the displayed continuum family; at equal rate, the standing-amplitude class and the remaining continuation witnesses separate structures whenever an admitted standing witness differs;
4. the newly activated coordinate class is *Conjugation/LNV*;
5. this coordinate is fibrewise independent of the retained oscillation packet;
6. D2 proves neither a common generator with oscillation continuation nor a global product factorization.

Proof. The first and third items are [theorem 14.1](#); exact separator existence for arbitrary distinct enriched classes is SCIC detectability. The universal witness is a physical $\Delta L = 2$ response, so the coordinate class is conjugation/LNV rather than absolute metric. The marked affine subfamily lies inside $\mathcal{V}_{76}^{\text{tr}}(H)$; [theorem 6.6](#) proves channel-by-channel constancy of the entire in-situ base packet while the D2 response varies, proving fibrewise independence. A converse factorization of all D2 witnesses and source incidences has not been proved and is not assumed. $\square$

14.2 Standing-coordinate incidence data

Ledger field	D2 result
Target-complete identifier	$\widehat{T}_{D2}^{\text{Ge}}$, including checksum-locked ^{76}Ge response and SOE-D2-1.
Base image	$\text{im } \rho_{D2} = Q_{\text{osc}}^{\text{SCIC}}$.
Coarse fibre effect	ParameterizedMixed; at least continuum many components in every fibre.
Exact separator export	Physical rate on the affine family; exact nonempty standing-amplitude, kernel, continuation, and deletion separator family for arbitrary split pairs.
Coordinate class	Conjugation/LNV transition response.
Relation to oscillation standing	Fibrewise independent: D2 response varies while Ξ_{osc} is fixed; no common generator proved.
S/R/P/Y profile	Exact pointwise code from D2-07, with the correction amplitude fixed under deletion.
Failure boundary	Exact N/A/fail/undefined/ROOS/source-update/target-update partition from D2-11.
PCR-5E comparison tags	‘full-base-image’, ‘independent-coordinate’, ‘conjugation-LNV’, ‘parameterized-mixed’, ‘no-common-generator-yet’.

D1 and D2 have the same full base image but different newly visible coordinates: absolute metric support in D1 and conjugation/LNV response in D2. Equal base images do not identify the physical generators.

15 Pairwise and higher overlap closure

15.1 Solved chart maps and complete-core equalizers

Let

$$\mathfrak{B}_{D2} = \{\text{CMP, LAW, LOCAL, HID, COMP, SOL, HYB}\}.$$

For every solved chart point $x \in Z_{a,A}^{D2}$, define the concrete complete-core map

$$\begin{aligned} \mathcal{C}_{a,A}^{D2}(x) = & (p_a^{\text{base}}(x), [A_A(x)]_{D2}^{\text{St}}, \overline{\text{Resp}}_{76}(A_A(x)), \text{Stat}_A(x), \\ & \text{Att}_A(x), \text{Match}_A(x), \text{Cont}_A(x), \\ & \text{Comp}_A(x), \partial \text{Stat}_A(x), \text{High}_A(x)). \end{aligned}$$

Here A_A is the explicit family amplitude of [section 9.3](#); for the compiled chart, $A_{\text{CMP}} = A_H + C$, with $C \in \mathcal{V}_{76}^{\text{src}}$. On the marked affine witness only, this specializes to zA_{76}^* . Only the standing-amplitude class and descended complete response section enter terminal identity. Raw amplitudes and response-frame representatives remain framed provenance. The overlap is the homotopy equalizer of these complete-core maps, not of family labels or independently selected nuisance coordinates.

For a chart pair (A, B) , choose a base-core isomorphism α and one common complete response frame η , allowing transport by a specified arrow of $\Gamma_{76}^{\text{resp}}$. The exact ordered residual is

$$\begin{aligned} \mathcal{E}_{aA,bB,\eta}^{\text{ov}}(x, y, \alpha) = & (E_{\text{base}}(\alpha), E_{\text{frame}}(\eta), \Delta[A]_{A,B}^{\text{St}}, \Delta K_{A,B,\eta}, \Delta \Gamma_{A,B,\eta}, \Delta \text{Stat}_{A,B}, \\ & \Delta \text{Att}_{A,B}, \Delta \text{Match}_{A,B}, \Delta \text{Cont}_{A,B}, \Delta \text{Comp}_{A,B}, \Delta \partial \text{Stat}_{A,B}, \Delta \text{High}_{A,B}). \end{aligned} \quad (20)$$

Every difference is transported along α and the response-frame arrow. The first nonzero coordinate is the exact separator and emptiness certificate. A zero residual carries the complete-core isomorphism and diagonal isotropy action; cross-frame splicing is not an overlap.

15.2 Twenty-eight chart-pair systems

Fix one lawful complete response frame η . For the compiled band the raw solved map is

$$K_{\text{CMP},\eta}(H, C) = \mathcal{R}_{76,\eta}(A_H + C), \quad C \in \mathcal{V}_{76}^{\text{src}},$$

and for every intrinsic band use the fibred map $K_{F,\eta} = \mathcal{R}_{76,\eta}A_F$ from the solved source-amplitude formulas. The marked affine line $C = C_{H,z}^*$ is only a downstream cardinality witness inside this complete compiled chart and does not define any overlap domain. The displayed kernel equation in each pair system is evaluated at this same η ; the remaining coordinates include frame coherence and are those of [equation \(20\)](#).

Pair	Exact standing-amplitude/kernel equalizer	Additional complete-core equalities and first separators
CMP–CMP	$\mathcal{R}_{76,\eta}(A_H + C) = \mathcal{R}_{76,\eta}(A_{H'} + C')$, together with $[A_H + C]_{D2}^{\text{St}} = [A_{H'} + C']_{D2}^{\text{St}}$. No equality of raw C, C' is required unless an admitted standing witness detects it.	Same transported base core, response-frame orbit, process ownership, deletion consequence, status, continuation, failure and isotropy.
CMP–LAW	$\mathcal{R}_{76,\eta}(A_H + C) = \mathcal{R}_{76,\eta} \sum_{\pi} A_{\pi}^f L_{\pi} A_{\pi}^i$, with equality of the transported standing-amplitude classes.	LAW memory, order, matching, continuation, deletion and failure equal the compiled core.
CMP–LOCAL	$\mathcal{R}_{76,\eta}(A_H + C) = \mathcal{R}_{76,\eta}(\sum_{\pi} V_r G \cdots G V_1 + A^{\text{contact/ind}})$, with equality of the transported standing-amplitude classes.	Representation/rank, thresholds, induced terms, matching, deletion and failure.
CMP–HID	$\mathcal{R}_{76,\eta}(A_H + C) = \mathcal{R}_{76,\eta} \bigoplus_r P_{\text{vis}} \langle r V \Omega \rangle P_{\text{in}}$, with equality of the transported standing-amplitude classes.	Minimal hidden support, projection, memory/population, reduct, status and isotropy.
CMP–COMP	$\mathcal{R}_{76,\eta}(A_H + C) = \mathcal{R}_{76,\eta} \sum_{\pi} D_f B_{\pi} D_i$, with equality of the transported standing-amplitude classes.	Constituent retention, derivation, support, matching, deletion and failure.

Pair	Exact standing-amplitude/kernel equalizer	Additional complete-core equalities and first separators
CMP-SOL	$\mathcal{R}_{76,\eta}(A_H + C) = \mathcal{R}_{76,\eta} \sum_{\pi} \int \langle \phi_f \mathcal{O}_{\pi} \phi_i \rangle d\mu$, with equality of the transported standing-amplitude classes.	Profile/topology, persistence, endability, attachment, continuation and status.
CMP-HYB	$\mathcal{R}_{76,\eta}(A_H + C) = \mathcal{R}_{76,\eta} \bigoplus_{\lambda} \sum_{\pi} \text{Contr}_{\pi}(\{A_i\}, \{T_e\})$, with equality of the transported standing-amplitude classes.	Every higher relation, interference sector, glue, cone, deletion/source-composition and provenance.
LAW-LAW	$K_{\text{LAW},\eta}(\theta) = K_{\text{LAW},\eta}(\theta')$.	Law action, memory, matching, continuation, deletion, status and complete failure frontier.
LAW-LOCAL	$\mathcal{R}_{76,\eta} \sum_{\pi} A^f L_{\pi} A^i = \mathcal{R}_{76,\eta} (\sum_{\pi} V_r G \cdots G V_1 + A^{\text{ind}})$.	Thresholds, induced terms, oriented matching, continuation, deletion and failure.
LAW-HID	$K_{\text{LAW},\eta} = \mathcal{R}_{76,\eta} \bigoplus_r P_{\text{vis}} \langle r V \Omega \rangle P_{\text{in}}$.	Hidden support/projection, memory, matching, deletion and failure.
LAW-COMP	$K_{\text{LAW},\eta} = \mathcal{R}_{76,\eta} \sum_{\pi} D_f B_{\pi} D_i$.	Constituent/derivation support, matching, continuation, deletion and frontier.
LAW-SOL	$K_{\text{LAW},\eta} = \mathcal{R}_{76,\eta} \sum_{\pi} \int \langle \phi_f \mathcal{O}_{\pi} \phi_i \rangle d\mu$.	Profile/topology, persistence/endability, attachment, continuation and failure.
LAW-HYB	$K_{\text{LAW},\eta} = K_{\text{HYB},\eta}$ with the full decorated contraction.	Every higher incidence, interference sector, cone, source minima, deletion and provenance.
LOCAL-LOCAL	$K_{\text{LOCAL},\eta}(\theta) = K_{\text{LOCAL},\eta}(\theta')$ using full propagators and induced terms.	Representation, active rank, thresholds, matching, continuation, deletion and status.
LOCAL-HID	$K_{\text{LOCAL},\eta} = K_{\text{HID},\eta}$.	Local induced terms versus minimal hidden support/projection, thresholds, memory and failure.
LOCAL-COMP	$K_{\text{LOCAL},\eta} = K_{\text{COMP},\eta}$.	Constituent derivation, local rank/threshold, matching, continuation and deletion.
LOCAL-SOL	$K_{\text{LOCAL},\eta} = K_{\text{SOL},\eta}$.	Profile/topology, persistence/endability, local threshold, attachment and failure.
LOCAL-HYB	$K_{\text{LOCAL},\eta} = K_{\text{HYB},\eta}$.	Cross residuals, higher relations, interference sectors, cone and source-composition minima.
HID-HID	$K_{\text{HID},\eta}(\theta) = K_{\text{HID},\eta}(\theta')$ after lawful Stinespring transport.	Minimal support, dilation/projection, memory/population, deletion, status and isotropy.
HID-COMP	$K_{\text{HID},\eta} = K_{\text{COMP},\eta}$.	Hidden support versus constituents, projection/derivation, memory, matching and frontier.
HID-SOL	$K_{\text{HID},\eta} = K_{\text{SOL},\eta}$.	Hidden support versus profile support, memory/continuation, attachment, deletion and failure.
HID-HYB	$K_{\text{HID},\eta} = K_{\text{HYB},\eta}$.	Co-carriage/projection, higher relations, interference, cone and source minima.
COMP-COMP	$K_{\text{COMP},\eta}(\theta) = K_{\text{COMP},\eta}(\theta')$.	Constituents, binding, derivation, matching, deletion, status and isotropy.
COMP-SOL	$K_{\text{COMP},\eta} = K_{\text{SOL},\eta}$.	Constituent versus profile support, derivation, persistence/endability, attachment and frontier.
COMP-HYB	$K_{\text{COMP},\eta} = K_{\text{HYB},\eta}$.	Extra constituents/cross incidence, interference, cone, deletion minima and provenance.
SOL-SOL	$K_{\text{SOL},\eta}(\theta) = K_{\text{SOL},\eta}(\theta')$.	Profile, topology, persistence, endability, attachment, continuation, deletion and status.
SOL-HYB	$K_{\text{SOL},\eta} = K_{\text{HYB},\eta}$.	Higher relation, co-carriage, profile cross effect, cone, source minima and failure.

Pair	Exact standing-amplitude/kernel equalizer	Additional complete-core equalities and first separators
HYB–HYB	$K_{\text{HYB},\eta}(\theta) = K_{\text{HYB},\eta}(\theta')$ for the complete decorated contractions.	Every higher incidence, coherent/incoherent sector, glue, cone, deletion, status and isotropy.

These are all unordered band pairs with repetition: $7 \cdot 8/2 = 28$. Each row is instantiated for every pair of source skeletons in the corresponding bands. Because the amplitude maps are explicit finite contractions, the equalizer is an actual symbolic solution locus at PCR-5B scope rather than an instruction to “compare responses.”

15.3 Direct higher-overlap system

For a finite tuple $\mathbf{x} = (x_0, \dots, x_m)$ of solved chart points, pairwise nonemptiness is insufficient. The direct higher residual is

$$\Omega_{\mathbf{x}}^{D2} = (\Omega_{\text{frame}}, \Omega_{\text{cocycle}}, \{\Delta[A]_{0i}^{\text{St}}\}_{i>0}, \{K_i - U_{0i}K_0\}_{i>0}, \Omega_{\text{high}}, \Omega_{\text{del}}, \Omega_{\text{cont}}, \Omega_{\text{cone}}, \Omega_{\text{stat}}).$$

It checks one common base frame, cocycle coherence of core isomorphisms, equality of all transported standing-amplitude classes and response maps, every primitive higher source incidence, deletion/source-composition consequences, continuation and cone compatibility, and one total status. Raw amplitude representatives remain in framed provenance and are not required to agree unless their difference survives standing-amplitude descent. Its zero locus is the exact higher homotopy fibre product; a nonzero coordinate is a typed emptiness certificate.

Theorem 15.1 (D2-10: complete enriched overlap closure). *The twenty-eight pair systems of [equation \(20\)](#) and every finite direct higher system are exact. Their simplicial groupoid and codescent satisfy*

$$\mathcal{A}_{D2}^{\text{exec}} \simeq \text{codesc}(\mathcal{O}_{D2,\bullet}^{\text{exec}}) \simeq \mathcal{N}(\hat{T}_{D2}^{\text{Ge}}).$$

Consequently:

1. *source-distinct presentations with isomorphic complete enriched cores are counted once at SCIC resolution while framed provenance and isotropy remain available;*
2. *equal scalar rates do not identify unequal standing-amplitude classes, kernels, continuations, source-composition consequences, failure behavior, or higher incidence;*
3. *every lawful D2 survivor occurs in an executed chart and every chart point is sound;*
4. *the D2-05 classwise and global fibre quotients are semantic rather than registry based.*

Proof. D2-02 and the source-amplitude normal-form theorem give the source cover. D2-03 and the seven chart-classification theorems give solved subobjects and exact complements with explicit response maps. The twenty-eight pair systems are the literal complete standing-core equalizers of those maps, with raw amplitude first descended through $[J]_{D2}^{\text{St}}$. The higher residual adds the common cocycle and primitive higher-incidence data that pairwise equalizers cannot imply. Zero loci are therefore exactly the pairwise and higher homotopy fibre products. Enriched SCIC identifies precisely complete-core isomorphisms, so codescent neither over-identifies physical differences nor retains pure duplicate presentations. Coverage and soundness come from the semantic source and response theorems, not from the finite number of band names. $\square$

16 Failure, undefinedness, N/A, and update closure

16.1 Total status classification

For every raw lift X and every declared D2 relation j , applicability $A_j(X)$ is fixed from source and target typing before evaluation. The total status compiler uses the ordered decision tree:

$$\text{Stat}_j(X) = \begin{cases} \text{TU}, & \text{target identity or response primitive changes,} \\ \text{SU}, & \text{a new source primitive is required at fixed target,} \\ \text{NA}, & A_j(X) = 0 \text{ by theorem,} \\ \text{ROOS}, & A_j(X) = 1 \text{ but the fixed response cannot interpret the output,} \\ \text{Undef}, & A_j(X) = 1 \text{ and a required active quantity is unresolved or ill defined,} \\ \text{Fail}, & A_j(X) = 1 \text{ and the defined relation fails,} \\ \text{Pass}, & A_j(X) = 1 \text{ and the defined relation passes.} \end{cases}$$

The priority prevents an update from being disguised as inapplicability and prevents an active undefined response from being called zero.

16.2 Exact complement of each chart

For a raw chart $\Theta_a^{D2, \text{raw}}$, define disjoint subobjects

$$N_a^{D2}, F_a^{D2}, U_a^{D2}, O_a^{D2}, SU_a^{D2}, TU_a^{D2}, Z_a^{D2}$$

for N/A, fail, undefined, response-out-of-scope, source update, target update, and survivor status. Their union is the raw chart. Each non-survivor point retains the first decisive relation, complete failure trace, and reopening condition; a first-failure report is diagnostic but does not replace the full membership proof.

Table 4: Principal D2 failure and update frontiers.

Status	Typical D2 instance	Scientific meaning
N/A	A theoremmically absent SMEFT or LME relation on a different valid response chart	Relation does not apply; no pass or failure inferred
Failure	Defined rate/likelihood, gauge, vacuum, matching, or common-cone relation fails	Fixed raw source point is not a survivor
Undefined	Required NME/LEC/kernel/continuation or common witness is unresolved	Fail closed for survival; not physical zero
Response out of scope	Physical source output exists but the fixed EXP_RATE instrument cannot interpret its final-state topology	No empirical exclusion; response/target enlargement required
Source update	New source sort, incidence, constitutive law, activity law, or essential infinitary architecture	Reopen D2-02/PCR-3 scope
Target update	New isotope, data release, event topology, multi-isotope likelihood, or response primitive	New target specification and lift calculation

Theorem 16.1 (D2-11: exact complement and reopening theorem). *The status compiler is total, mutually exclusive, applicability preserving, and invariant under lawful source/response redescription. For every raw*

D2 chart,

$$\Theta_a^{D2, \text{raw}} = N_a^{D2} \sqcup F_a^{D2} \sqcup U_a^{D2} \sqcup O_a^{D2} \sqcup SU_a^{D2} \sqcup TU_a^{D2} \sqcup Z_a^{D2}.$$

The compiled prequotient pass object is

$$\tilde{Z}_{a, \text{CMP}}^{D2} = \left\{ (\theta, C, \eta) \left| \begin{array}{l} \theta \in Z_a, \quad C \in \mathcal{V}_{76}^{\text{src}}, \quad \eta \in \mathcal{E}_{76}^{\text{adm}}, \\ A_{a, F}(\theta) + C \in \mathcal{A}_{76}^{\text{phys}}, \\ C \in \mathcal{V}_{76}^{\text{tr}}(H_a(\theta)) \text{ for the compiled transparent chart,} \\ \mathcal{R}_{76, \eta}(A_{a, F}(\theta) + C) \text{ is defined,} \\ \text{Acc}_{90, \eta}(\mathcal{R}_{76, \eta}(A_{a, F}(\theta) + C)) = 1 \end{array} \right. \right\}.$$

Its exact complement inside $Z_a \times \mathcal{V}_{76}^{\text{src}} \times \mathcal{E}_{76}^{\text{adm}}$ is the union of source-parameter failure, frame failure, undefined/ROOS response, and acceptance failure, together with the inherited base and source-extension complements. A source presentation with an activated old-target incidence is not relabelled as a failed transparent correction: it is routed by its in-situ retained packet to the corresponding PCR-5B core when one exists, or to TU when the old target/interface itself changes. The marked affine subchart

$$\{(\theta, C_{H_a(\theta), z}^*, \eta_\star) : \theta \in Z_a, \quad z \in I_{76, \eta_\star}^{\text{acc}}\}$$

lies in this survivor object. A base fibre is empty only if every raw lift in every lawful frame lies in a non-survivor component; D2-03 supplies the marked subchart over every base class.

Proof. D2-01 fixes branch applicability and update semantics before evaluation. Classical case distinction along the ordered compiler gives totality and disjointness. Lawful arrows preserve typing and all physical response relations, hence preserve status. D2-02 covers every raw lift, so the partition is exact. Fibre emptiness is equivalent to absence of survivor components in all frames and charts. D2-03 supplies a survivor in the executed compiled chart over each base class, defeating emptiness. $\square$

No present result turns a null observation into exact lepton-number conservation or a physical zero theorem. Conversely, no undefined or out-of-scope response is counted as zero or as an accepted lift.

17 Joint-target interface and PCR-5E input

17.1 Exact enriched quotient

Combining the retained-transparency theorem with D2-04, D2-05, D2-08, and D2-10 gives

$$\boxed{Q(\hat{T}_{D2}^{\text{Ge}}) \cong \bigsqcup_{q \in Q_{\text{osc}}^{\text{SCIC}}} \pi_0 \mathcal{L}_{D2}(q)}, \quad \boxed{\text{im } \rho_{D2} = Q_{\text{osc}}^{\text{SCIC}}}, \quad \boxed{|\pi_0 \mathcal{L}_{D2}(q)| \geq \mathfrak{c} \quad \forall q \in Q_{\text{osc}}^{\text{SCIC}}}.$$

The quotient is an exact mixed discrete, continuous, functional, and categorical moduli object at the inherited symbolic scope. The displayed continuum is a lower bound from the compiled physical-response family, not an exact upper-cardinality theorem.

17.2 D2 marginal datum

Define

$$\mathbf{H}_{D2} := \left(\hat{T}_{D2}^{\text{Ge}}, \text{SOE-D2-1}, \mathcal{E}_{76}^{\text{adm}}, \Gamma_{76}^{\text{resp}}, \{\mathcal{R}_{76,\eta}, \mathfrak{G}_{76,\eta}, \text{Acc}_{90,\eta}\}_{\eta}, \text{im } \rho_{D2}, \{\mathcal{L}_{D2}(q)\}_{q \in Q_{\text{osc}}^{\text{SCIC}}}, \right. \\ \left. \mathcal{A}_{D2}^{\text{exec}}, \Xi_{D2}, \text{Sep}_{D2}^{\text{rel}}, \text{Comp}_{D2}, \right. \\ \left. \text{Ov}_{D2}, \partial \text{Stat}_{D2}, \text{Prov}_{D2}, \Sigma_{D2}^{\text{new}} \right).$$

It contains the fixed target/source identifier, complete response-frame groupoid and fibred response maps, survival quantifier, base image, connected-class fibres, global atlas, standing packet, relative separator families, source-composition hypergraphs, pairwise and higher overlap nerve, status and update frontier, framed provenance, and standing-coordinate diagnostic.

Theorem 17.1 (D2-12: exact joint-handoff theorem). *The datum $\mathbf{H}_{D2}$ is sufficient for PCR-5E to form the D2 marginal of the independently generated joint target without reconstructing omitted D2 source, response, restriction, chart, overlap, source-composition, or failure information. Its diagnostic entry is*

$$\Sigma_{D2}^{\text{new}} = \text{Conjugation/LNV},$$

with full base image, ParameterizedMixed fibres over every q , fibrewise independence from the retained oscillation packet, and no established common generator with oscillation or D1.

Proof. D2-01 through D2-11 construct every component of $\mathbf{H}_{D2}$ and prove invariance under lawful source and response presentation. PCR-5C's joint-target theorem requires exactly the marginal cores, source/provenance frames, restriction maps, compatibility incidences, separator families, composition data, and failure/update frontiers represented here. The coordinate label is diagnostic only. No marginal tuple is promoted to a joint survivor: PCR-5E must independently generate and evaluate one common source point. $\square$

17.3 Physical interpretation

The present result is structurally parallel to D1 in its action on the base quotient and physically different in its new coordinate. Absolute kinematic scale and $\Delta L = 2$ transition response can both vary while the entire oscillation standing packet remains fixed. Therefore neither selects the oscillation carrier architecture at the amplitude target scopes.

D2 adds two calibrations. First, the present LNV null search refines possible responses and excludes parts of their parameter space without instantiating an LNV event. Second, source-block necessity is pointwise: the compiled amplitude or an intrinsic LNV block may be identity essential, target-survival essential, redundant, partially synergistic, or synergistic according to D2-07. A zero net response is not identified with absence of LNV source structure.

The later joint question is whether D3–D5 couple absolute metric, conjugation/LNV, closure/unitarity, transport/population, and symmetry/charge through one common invariant or a genuinely higher-arity source relation. That remains a PCR-5E theorem.

18 Robustness and falsifiability

18.1 Dependency of the principal result

The endpoint depends on the ordered implications

$$\begin{aligned}
& \text{target freeze, fibred response, and one-frame survival} \implies \text{D2-01,} \\
& \text{PCR-3 coverage, exact source-amplitude normal form, and the conservative SOE-D2-1 extension} \implies \text{D2-02,} \\
& \text{exact transition-amplitude realization, complete source domain, and additive compatibility} \implies \text{exact physical process-to-response,} \\
& \text{constraint-preserving physical amalgamation, activity-reduced trace, marked affine witness, exact reduct, and residual inheritance} \implies \text{D2-03,} \\
& \text{retained standing witnesses} \implies \text{D2-04 no merger,} \\
& \text{six intrinsic maps and the complete } C\text{-indexed compiled chart} \implies \text{D2-05 fibres,} \\
& \text{no-retuning deletion and enriched SCIC} \implies \text{D2-07 S/R/P/Y,} \\
& \text{one marked survivor over every B15 frame} \implies \text{im } \rho_{D2} = Q_{\text{osc}}^{\text{SCIC}}, \\
& \text{accepted real interval and rate standing} \implies |\pi_0 \mathcal{L}_{D2}(q)| \geq \mathfrak{c} \, \forall q, \\
& \text{28 equalizers and direct higher residual} \implies \text{D2-10 codescent.}
\end{aligned}$$

The endpoint would require revision if a fixed-target counterexample defeated one of these implications, if the candidate domain were incomplete, if a standing-bearing coordinate were omitted, if the proposed extension replaced rather than retained the base target, or if any restriction, lift, split, overlap, or joint-target map were ill typed.

18.2 Adversarial tests

ID	Adversarial test	Disposition
H01	Treat the present null search as an exact zero-amplitude or lepton-number-conservation theorem.	Rejected by the nonempty accepted positive-rate locus and Proposition 3.2.
H02	Use $m_{\beta\beta} = \sum_i U_{ei}^2 m_i $ as the universal D2 carrier variable.	Rejected: the formula is restricted to the LME specialization; EXP_RATE is the primary mechanism-neutral branch.
H03	Turn NME, detector, calibration, or statistical variation into a carrier selector.	Rejected unless it changes a descended target-authorized physical consequence; otherwise it lies in the response groupoid or nuisance descent.
H04	Form a D2 candidate only after inspecting its likelihood or desired base class.	Rejected by the response-mutation invariance theorem; raw intrinsic and compiled source domains are formed before response evaluation.
H05	Attribute SOE-D2-1 silently to fixed PCR-3.	Rejected: the conservative source extension, exact reduct, source sorts, and reopening boundary are declared explicitly.
H06	Promote an arbitrary target kernel to a physical source process.	Rejected: source recursion first produces a typed exact amplitude in $\mathcal{V}_{76}^{\text{src}}$; physical realization is licensed only on the contraction locus $\mathcal{A}_{76}^{\text{phys}}$.
H07	Claim that unitary dilation proves physicality without support, kinematic, covariance, charge, or owner typing.	Rejected: those homogeneous source equations define $\mathcal{V}_{76}^{\text{src}}$ independently, and the colligation theorem proves their transport to the defect blocks.
H08	Treat an infinitesimal interaction generator as the exact finite transition amplitude.	Rejected: the load-bearing object is the exact amplitude A , and $P_f \mathbb{U}_{76}(A) P_i = A$ holds literally. No derivative or small-time approximation enters the proof.

ID	Adversarial test	Disposition
H09	Suppress the intrinsic amplitude when adding the compiled correction.	Rejected: $A_{H,C}^{\text{tot}} = A_H + C$ is the complete source amplitude; the marked correction cancels it only by the explicit identity $A_H + (-A_H + zA_\star) = zA_\star$.
H10	Define the raw compiled domain by the response-visible direction $A_\star$.	Rejected: the raw domain is all $C \in \mathcal{V}_{76}^{\text{src}}$; $A_\star$ is selected later only as a marked survivor/cardinality witness.
H11	Omit generated correction amplitudes outside the marked affine line.	Rejected: D2-02 and the CMP chart include the entire C -domain and assign every point an exact physicality and response status.
H12	Recompute the correction after deleting a base block.	Rejected by D2-07: the original C is held fixed, so deletion exposes the exact no-repair mismatch.
H13	Infer a defined zero response from deletion or undefinedness.	Rejected: the zero-amplitude point is the defined complete source point $C = -A_H$; undefined/ROOS/N/A remain separate statuses.
H14	Identify exact-amplitude equality with raw source-provenance equality.	Rejected: terminal identity uses the fibred standing-amplitude quotient and complete response/continuation signature; raw amplitude remains framed provenance.
H15	Merge standing-distinct amplitudes because their present scalar rates agree.	Rejected: kernel, continuation, attachment, matching, deletion, failure, and higher-incidence witnesses remain in enriched SCIC.
H16	Refine SCIC by an amplitude norm, preferred representative, source label, or naturalness score.	Rejected by witness completeness and the no-second-classifier theorem.
H17	Splice a nuclear map, detector model, and statistical verdict from different response frames.	Rejected by one-frame survival and lawful response-groupoid descent.
H18	Replace response-frame existence by a post-hoc best-fit choice.	Rejected: the existential complete-frame semantics is fixed in D2-01 before candidate evaluation.
H19	Count family names or source files as semantic exhaustion.	Rejected: each family has an explicit exact-amplitude map, guarded residual, solved groupoid, complement, and overlap datum.
H20	Use pairwise nonemptiness to infer higher overlap coherence.	Rejected: D2-10 evaluates a direct finite higher residual with common framing, cocycles, higher incidence, deletion, continuation, cone, and status.
H21	Collapse distinct base SCIC classes in the richer target.	Rejected by retained base witnesses and PCR-5C no merger.
H22	Eliminate a base class because one presentation fails D2.	Rejected: elimination requires exhaustion of every framed provenance presentation and every source-lawful D2 completion over the class.
H23	Interpret the universal compiled chart as a claim that every microscopic theory contains the correction.	Rejected: it is a declared target-complete comparison component with exact base reduct, not a statement about every intrinsic presentation.
H24	Demand a local renormalizable electroweak-QFT realization of every target-scale process.	Rejected as a comparison-class change; intrinsic local-QFT charts retain all black-box-induced consequences.
H25	Exclude the marked family as fine tuned or unnatural.	Rejected: the family is source generated before target evaluation, and naturalness has no standing-selector authority.
H26	Use the exact unitary colligation as a unique microscopic ontology.	Rejected: it is one exact target-scale physical realization of the admitted amplitude, not a UV-uniqueness theorem.
H27	Treat C itself as a standalone physical contraction.	Rejected: only the complete amplitude $A_H + C$ is physically realized; C is a typed correction coordinate.

ID	Adversarial test	Disposition
H28	Replace exact acceptance by the marked continuum subinterval.	Rejected: $I_{76,\eta_*}^{\text{acc}}$ is the exact pass locus; I_{76,η_*}^* is only a cardinality witness.
H29	Claim an exact upper cardinal from the displayed real interval.	Rejected: intrinsic symbolic, functional, categorical, provenance, and isotropy moduli remain admitted.
H30	Infer Majorana character or nonzero LNV realization from the D2 standing coordinate.	Rejected: D2 identifies a conjugation/LNV response coordinate and includes a defined zero-amplitude survivor.
H31	Infer a global product decomposition from fibrewise independence.	Rejected: only retained-base independence and full base image are proved; joint compatibility remains for PCR-5E.
H32	Replace the joint D1–D5 target by an intersection of marginal images.	Rejected by PCR-5C one-point joint-target discipline and the D2-12 handoff.
H33	Retain a stale infinitesimal-generator or optional-dilation source signature.	Rejected: the fixed core signature uses Corr_{76} , complete-amplitude physicality, and RealAmp_{76} ; the exact colligation is the realization theorem itself.
H34	Alter response maps or acceptance and thereby change the raw source domain.	Rejected by the direct no-backflow mutation: only the later survivor partition may change.
H35	Let a source-coherent but noncontractive total amplitude pass as a physical process.	Rejected by the explicit residual $E_{\text{amp}} = 0 \iff A \in \mathcal{A}_{76}^{\text{phys}}$.
H36	Use amplitude-level cancellation to infer exact process cancellation.	Rejected because cancellation occurs at the exact transition-amplitude object before realization, and the observed unitary block equals that amplitude exactly.
H37	Claim exact transition realization for a noncontraction.	Rejected: the raw point remains source generated but receives physical-amplitude failure; the theorem realizes only $\mathcal{A}_{76}^{\text{phys}}$.
H38	Challenge the marked continuum by changing the empirical target or response-frame semantics.	This is a target update and reopens D2-01/D2-03; it is not a same-target counterexample.
H39	Treat the intrinsic and correction amplitudes as two independently realized scattering processes and multiply or probabilistically mix them.	Rejected: the compiled source equation forms $A_H + C$ before realization, and one unitary colligation has that sum as its literal transition block; theorem 6.9 gives the exact interference response.
H40	Replace the retained physical base carrier by the aggregate amplitude colligation and recover H only as formal provenance.	Rejected by the constraint-preserving source-faithful theorem: $W_{H,C}$ is a split-anchored physical extension with activity-reduced trace $P_{H,C}^{\text{red}}$, exact D2 transition $A_H + C$, physical restriction W_H , a semantic common cone, and natural physical deletion.
H41	Treat the source-faithful construction as a product or intersection of marginal survivor verdicts.	Rejected: the split attachment is formed before response evaluation and lemma 6.3 proves that every inherited, pure-process, cross, and common-cone residual vanishes on one semantic physical model.
H42	Delete the correction only syntactically while altering the physical base carrier.	Rejected: the declared splitting gives a physical retraction that is the identity on every base sort, operator, parameter, incidence, and standing witness.
H43	Obtain the correct amplitude while failing to realize the activity-reduced compiled source trace.	Rejected: the base cells occur once and (7) gives $\chi_{\text{car}}(W_{H,C}) \simeq P_{H,C}^{\text{red}}$; a zero correction is stripped rather than retained as false multiplicity.
H44	Let the formal source reduct be unrelated to physical restriction.	Rejected by the natural commuting square $\chi_{\text{car}} \circ \text{Del}_{c_{76}}^{\text{phys}} \simeq U_{D2}^{\text{CMP}} \circ \chi_{\text{car}}$; the source reduct is the carrier-trace shadow of the physical retraction.

ID	Adversarial test	Disposition
H45	Put every named witness on one pushout but allow matching, continuation, vacuum/deformation, causal, induced-effect, or higher-incidence equations to fail.	Rejected by the explicit cross equations (9)–(16) and the split residual decomposition; the common apex lies inside $\text{PhysMod}_{4,D2}^{\text{sem}}$, not merely in the ambient diagram category.
H46	Retain a source-null zero correction as an active carrier block or use it to inflate the continuum count.	Rejected by (7); the trace is H at $C = 0$, and the continuum theorem uses $I_{76,H}^{\star,\text{act}}$, obtained by removing at most one null point.
H47	Recover the base only after deleting the active D2 correction while allowing the correction to change a fixed oscillation standing coordinate when present.	Rejected by theorem 6.6 : Φ_4^{ret} and all seven channels are evaluated in situ and equal the base values before correction deletion.
H48	Add a compiled correction with an activated old-target incidence and count it as a failed lift over the original base class.	Rejected by corollary 6.7 : such a source is routed by its retained packet to the appropriate base core or to target update; it is not silently misclassified in the transparent compiled chart.

18.3 Direct falsifiers

The capstone reopens if any of the following is established at the fixed target and comparison scope:

1. a PCR-5B/B15 source point whose complete D2 transition use does not factor through the finite exact-amplitude normal form;
2. a source-active exact amplitude omitted from $\mathcal{V}_{76}^{\text{src}}$, or a member of $\mathcal{V}_{76}^{\text{src}}$ whose typing equations are not source authorized;
3. a contraction $A \in \mathcal{A}_{76}^{\text{phys}}$ for which the displayed colligation is not unitary or its observed transition block is not exactly A , a compiled source for which the complete amplitude does not equal $A_H + C$, or a framed physical base realization for which $W_{H,C}$ fails to realize $P_{H,C}^{\text{red}}$, restrict physically to W_H , or belong to $\text{PhysMod}_{4,D2}^{\text{sem}}$;
4. a split-attachment failure: a nonclosed boundary identification, a carrier trace different from $P_{H,C}^{\text{red}}$, loss of the base retraction, or any nonzero inherited, pure-process, matching, owner, continuation, vacuum/deformation, causal, induced-effect, higher-incidence, or common-cone residual;
5. a framed base source for which the correction $-A_H + zA_\star$ is not in the independently generated raw correction domain;
6. failure of the complete-amplitude identity $A_H + (-A_H + zA_\star) = zA_\star$ on one physical source point;
7. absence of any source-authorized response-visible pair $(A_\star, \eta_\star)$ with positive finite rate;
8. failure of the one-frame response covariance, acceptance descent, or nondegenerate accepted neighborhood at $z = 0$;
9. a generated D2 survivor absent from every exact chart or a duplicate presentation not resolved by the complete overlap systems;
10. a standing-bearing amplitude distinction invisible to the declared enriched witness algebra, or an inert raw-amplitude distinction retained as terminal identity;
11. a zero correction whose activity-reduced trace still contains an active c_{76} block, or a base frame for which deleting the at-most-one null point destroys the continuum witness;

12. a compiled physical point for which the in-situ retained evaluator changes any fixed seven-channel oscillation coordinate while the correction is present, defeating [theorem 6.6](#);
13. a faithful same-target D2 lift whose restriction changes a fixed base separator or whose exact source reduct is not a PCR-5B/B15 source point.

A new isotope, analysis release, detector primitive, response category, or positive signal is a change of target rather than a counterexample to the stated fixed-target theorem.

19 Conclusion

PCR-5D2 extends the exact oscillation standing-carrier quotient by the fixed ^{76}Ge lepton-number-violating search response. The analysis separates five levels: source-generated exact transition amplitudes, exact-amplitude process realization, source-faithful attachment to the retained physical carrier, fibred nuclear and detector response, and statistical acceptance. For every contraction $A \in \mathcal{A}_{76}^{\text{phys}}$, the Julia–Halmos colligation is unitary and has observed transition block exactly A ; no infinitesimal generator is identified with a finite response.

The compiled source universe is parameterized by the complete response-independent correction space $\mathcal{V}_{76}^{\text{src}}$. For a framed base realization W_H , the total amplitude is $A_H + C$. On the contraction locus, the physical amalgamation $W_{H,C}$ has carrier trace $P_{H,C}^{\text{red}}$, D2 transition block $A_H + C$, and physical restriction W_H . The inherited, pure-process, cross, and common-cone residuals vanish separately, so the common apex is a simultaneous semantic model rather than a collection of compatible marginal data. The retained-oscillation theorem evaluates the old target on this same enriched object and proves channel-by-channel equality of Φ_4^{ret} with Φ_4 and of $\Xi_{\text{osc}}^{\text{ret}}$ with Ξ_{osc} while the correction remains present.

A response-visible amplitude $A_\star$, selected only after the source domain is fixed, determines

$$C_{H,z}^\star = -A_H + zA_\star, \quad A_{H,z}^{\text{tot}} = zA_\star.$$

The exact transition, event kernel, and rate are evaluated on one complete physical point. A nondegenerate accepted interval therefore supplies continuum many enriched standing classes over every base class. The intrinsic families, correction chart, source-composition strata, response descent, failure frontiers, and pairwise and higher overlaps are all resolved at the inherited symbolic scope.

Consequently,

$$\boxed{\text{im } \rho_{D2} = Q_{\text{osc}}^{\text{SCIC}}}, \quad \boxed{|\pi_0 \mathcal{L}_{D2}(q)| \geq \mathfrak{c} \quad \forall q \in Q_{\text{osc}}^{\text{SCIC}}}.$$

D2 does not select a proper subset of oscillation carriers. It refines every base class through a standing-visible conjugation/LNV response coordinate. The result does not force a nonzero LNV amplitude, Majorana character, light-Majorana exchange, a global product decomposition, a common generator with oscillation continuation, a unique carrier, or an absolute upper cardinal. Those remain separate questions for the later target extensions and the joint PCR-5E analysis.

A Notation, types, and status semantics

Symbol	Meaning
T_{osc}	Fixed neutrino-oscillation target-complete object.
$Q_{\text{osc}}^{\text{SCIC}}$	Exact PCR-5B oscillation SCIC quotient.

Symbol	Meaning
$\hat{T}_{D2}^{\text{Ge}}$	Target-complete D2 object: retained oscillation target, fixed ^{76}Ge response, corrected source comparison class, and complete D2 standing data.
SOE-D2-1	Conservative D2 source extension supplying the compiled target-scale transition-process family.
$W_H, B_{76}(W_H)$	Chosen semantic PCR-4/B15 physical representative and its complete retained split anchor.
$D_{76}(H, C)$	D2-only cell diagram carrying the coloured envelope, correction, exact-amplitude process, ownership, continuation, induced effects, failure maps, and higher incidences.
$W_{H,C}$	Constraint-preserving source-faithful split-anchored physical extension obtained by identifying the D2 cell boundary with the retained anchor of W_H ; it lies in $\text{PhysMod}_{4,D2}^{\text{sem}}$.
$\Phi_4^{\text{ret}}, \Xi_{\text{osc}}^{\text{ret}}$	In-situ retained PCR-4 evaluator and seven-channel oscillation standing packet evaluated on the enriched object while D2 cells remain present.
$\mathcal{V}_{76}^{\text{tr}}(H)$	Base-transparent compiled correction locus: physically admissible corrections whose in-situ retained packet equals that of W_H . For the fixed compiled chart it equals the full physically admissible correction locus.
$\text{PhysMod}_{4,D2}^{\text{sem}}$	Semantic simultaneous physical-model locus for the complete active PCR-4/D2 relation theory.
$\text{CarrierSurv}_{\text{sem}}(\hat{T}_{D2}^{\text{Ge}})$	Defined accepted response-frame survivor subcategory of $\text{PhysMod}_{4,D2}^{\text{sem}}$.
$\text{Res}_{\text{osc}}, \text{Del}_{c_{76}}^{\text{phys}}, U_{D2}^{\text{CMP}}$	Physical base restriction, no-retuning physical correction deletion, and its carrier-trace shadow at source level.
$\text{BB}_{76}(H)$	Colored process envelope over framed PCR-3/B15 source H .
$\mathcal{A}_{76}^{\text{src}}(H), \mathcal{V}_{76}^{\text{src}}$	Complete intrinsic D2 exact coherent transition amplitude of H and the complete source-defined amplitude vector space; the physical locus is the separate contraction set $\mathcal{A}_{76}^{\text{phys}}$.
$C, C_{H,z}^*$	General compiled source parameter $C \in \mathcal{V}_{76}^{\text{src}}$, and the downstream marked witness $-\mathcal{A}_{76}^{\text{src}}(H) + zA_{76}^*$.
$P_{H,C}, P_{H,C}^{\text{red}}, P_{H,z}^*$	General raw compiled source, its pointwise activity-reduced carrier trace, and the marked affine witness. At $C = 0$, $P_{H,0}^{\text{red}} \simeq H$; $\text{RealAmp}_{76}(\mathcal{A}_{76}^{\text{src}}(H) + C)$ exists only on the contraction locus.
$\mathcal{E}_{76}^{\text{adm}}, \Gamma_{76}^{\text{resp}}$	Complete response-frame object and its lawful nuclear/detector/statistical groupoid.
$\mathcal{R}_{76,\eta}$	Fibred physical response map $\mathcal{D}_{76}(\eta_{\text{det}}) \circ \mathcal{N}_{76}(\eta_{\text{nuc}})$.
$\mathfrak{G}_{76,\eta}$	Positive Hermitian response form in frame η ; its diagonal is the physical rate.
(A_{76}^*, η_*)	Downstream response-visible exact-amplitude/frame witness selected only after the complete raw source domain is fixed, and before the observed likelihood bound is used.
$I_{76,\eta_*}^{\text{acc}}$	Exact accepted locus at η_* : $\{z : \text{Acc}_{90,\eta_*}(zK_{\star,\eta_*}) = 1\}$.
I_{76,η_*}^*	Marked continuum subinterval $[0, \delta_{76}^*] \subset I_{76,\eta_*}^{\text{acc}}$; $I_{76,H}^{*,\text{act}}$ removes the at-most-one source-null correction point when a nonnull compiled block is required.
U_{D2}^{BB}	Exact compiled-source reduct $\text{ActRed}_{T_{\text{osc}}} \circ \text{Forget}_{D2/T_{\text{osc}}}$.

Symbol	Meaning
$\mathcal{N}(\hat{T}_{D2}^{\text{Ge}}, Q(\hat{T}_{D2}^{\text{Ge}}))$	Concrete D2 standing-core groupoid and its connected-component quotient.
R_{D2}, ρ_{D2}	Standing-core restriction and induced map on SCIC components.
$\mathcal{L}_{D2}(q)$	Connected-subgroupoid homotopy lift fibre over base class q .
$\mathbb{K}_{76}$	Mechanism-neutral target-typed ^{76}Ge transition-kernel space.
$\Gamma_{1/2}^{0\nu}$	Physical inverse-half-life/rate coordinate.
Γ_{90}^{Ge}	Published combined 90% rate upper bound, $(1.9 \times 10^{26} \text{ yr})^{-1}$.
CMP	Exact SOE-D2-1 compiled chart band.
NA	Relation theoremtically inapplicable before evaluation.
Pass, Fail	Defined applicable relation passes or fails.
Undef	Applicable relation is unresolved or ill defined; fail closed for survival.
ROOS	Physical output exists but the fixed response cannot interpret it.
SU, TU	Source-comparison or target-specification update.

B Empirical files and integrity checks

B.1 Data inventory

The supplementary archive contains the five original public CSV files associated with the LEGEND record [26]. The integrity table `DATA_SHA256.csv` records local SHA-256 values, while the official MD5 values are reproduced in [section 3](#). The derived-data table records row counts, exposure totals, detector-label count, and event-window checks.

B.2 Profile likelihood and posterior objects

The released profile likelihood has 1000 rows. The LEGEND-only and combined posterior files have 179 and 185 rows, respectively. Their coordinate is an inverse-half-life/rate variable supplied by the collaboration release. The figures are direct plots of those released numerical arrays. They are used to verify target identity and illustrate the likelihood geometry, not to reconstruct collaboration confidence belts.

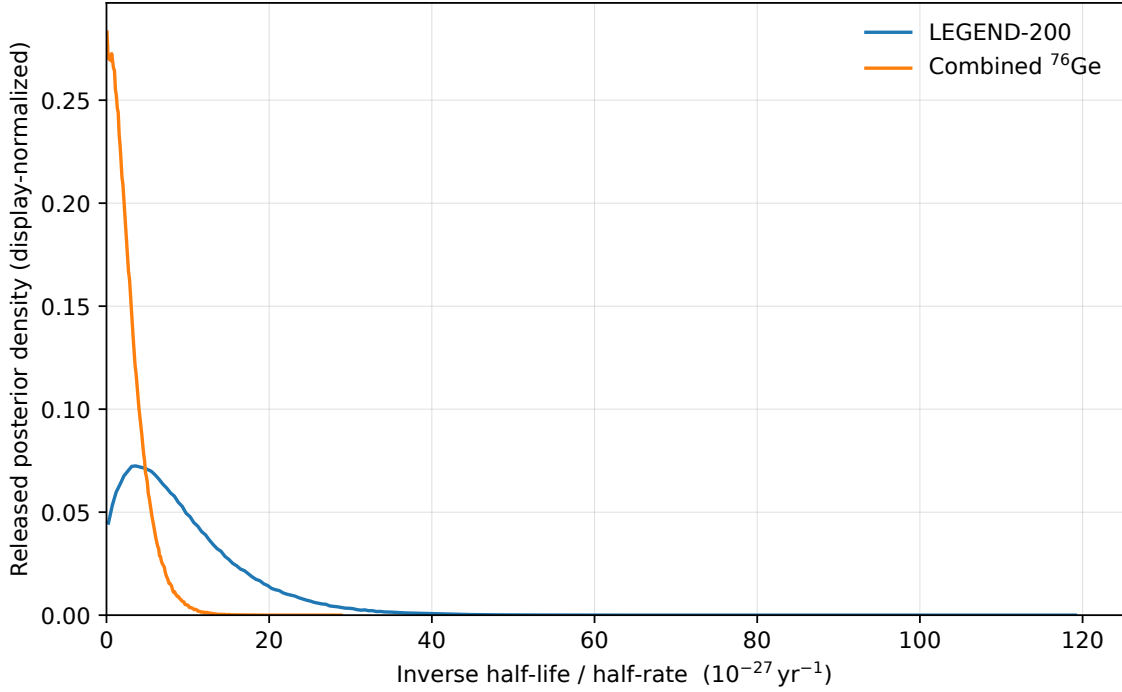


Figure 2: Released LEGEND-only and combined ^{76}Ge rate-posterior arrays, normalized for visual comparison by the included plotting script. The scientific target remains the collaboration’s released objects and stated inference semantics.

B.3 Event-window consistency check

The event file is preserved exactly. Applying the primary-paper gamma-line exclusion rule to the byte-complete event energies yields 11 retained rows from 13 released rows. The consistency check records the discrepancy between one prose energy and the exact file. Any future collaboration replacement of the record is a target update.

C Mechanism-neutral response derivations

C.1 Fibred source-to-record map

For every complete frame $\eta = (\eta_{\text{nuc}}, \eta_{\text{det}}, \eta_{\text{stat}})$,

$$\mathcal{R}_{76,\eta} = \mathcal{D}_{76}(\eta_{\text{det}}) \circ \mathcal{N}_{76}(\eta_{\text{nuc}}).$$

A lawful response-frame arrow $g : \eta \rightarrow \eta'$ supplies U_g with

$$U_g \mathcal{R}_{76,\eta} = \mathcal{R}_{76,\eta'}, \quad \mathfrak{G}_{76,\eta'}(U_g K_1, U_g K_2) = \mathfrak{G}_{76,\eta}(K_1, K_2).$$

The instrument, likelihood, and acceptance transform compatibly. This is the concrete response-descent datum used in D2-01, D2-05, D2-06, and D2-10.

C.2 Positive Hermitian rate form

In a finite amplitude chart $K \leftrightarrow (\mathcal{A}_i)_i$,

$$\mathfrak{G}_{76,\eta}(K_1, K_2) = \sum_{i,j} \mathcal{A}_i(K_1)^* G_{ij}(\eta) \mathcal{A}_j(K_2), \quad G(\eta) = G(\eta)^\dagger \geq 0.$$

Hence $\Gamma_{1/2}^{0\nu}(K, \eta) = \mathfrak{G}_{76,\eta}(K, K) \geq 0$. Functional or categorical charts use the corresponding positive kernel/integral form; the Hermitian covariance statement is unchanged.

C.3 Light-exchange and high-scale specializations

The light-exchange formula is evaluated only on its certified locus and with the target-authorized short-range contribution. The high-scale EFT formula is evaluated only under its SMEFT matching hypotheses. A source outside either specialization remains in EXP_RATE when its mechanism-neutral event kernel is defined.

C.4 Exact survival semantics

For an exact transition amplitude A ,

$$\text{Surv}_{76}^{\text{resp}}(A) = 1 \iff \exists[\eta] \in |\mathcal{E}_{76}^{\text{adm}}/\Gamma_{76}^{\text{resp}}| : \mathcal{R}_{76,\eta}(A) \text{ defined and } \text{Acc}_{90,\eta}(\mathcal{R}_{76,\eta}(A)) = 1.$$

The complete frame is the quantification unit. Profiling/marginalization is internal to η_{stat} ; no nuclear/detector/statistical splicing across inequivalent frames is allowed.

D Formal SOE-D2-1 source calculus

D.1 Core source signature

The conservative source extension has the core signature

$$(\mathcal{V}_{76}^{\text{src}}, \mathcal{A}_{76}^{\text{phys}}, \text{Corr}_{76}, \text{BB}_{76}, \text{Own}_{76}, \text{Par}_{76}, \text{RealAmp}_{76}, \text{D}_{76}(\cdot, \cdot), W_{\cdot, \cdot}, \text{Res}_{\text{osc}}, \text{Del}_{c_{76}}^{\text{phys}}, U_{D2}^{\text{CMP}}).$$

The source-defined correction space $\mathcal{V}_{76}^{\text{src}}$ is formed without nuclear maps, detector maps, response frames, likelihoods, acceptance values, standing classes, or endpoint information. For a B15 source frame H and $C \in \mathcal{V}_{76}^{\text{src}}$,

$$P_{H,C} = \text{Par}_{76}(H, \text{Own}_{76}(\text{BB}_{76}(H), \text{Corr}_{76}(C))), \quad A_{H,C}^{\text{tot}} = \mathcal{A}_{76}^{\text{src}}(H) + C.$$

The complete raw compiled source groupoid contains all such points. Candidate formation does not require $A_{H,C}^{\text{tot}} \in \mathcal{A}_{76}^{\text{phys}}$; that condition belongs to physical evaluation.

D.2 Source equations and exact reduct

The source equations require:

1. PCR-3/B15 source coherence of H ;
2. membership $C \in \mathcal{V}_{76}^{\text{src}}$ and exact amplitude composition $A_{H,C}^{\text{tot}} = \mathcal{A}_{76}^{\text{src}}(H) + C$;
3. the source activity law $\alpha_{76}(C) = 1 \iff C \neq 0$, with all correction-derived incidences stripped at $C = 0$;

4. ownership and boundary attachment of the correction presentation to the same enriched source point;
5. preservation of every primitive and higher source incidence under the colored envelope;
6. correction-block deletion with no retuning of C ;
7. the exact reduct

$$U_{D2}^{\text{CMP}}(P_{H,C}^{\text{red}}) = H.$$

Forgetting the correction commutes with lawful source arrows and returns the fixed B15 framing. D2-opacity means that the correction presentation has no retained oscillation action; retained base equality is proved downstream by the exact reduct and retained evaluator, not inserted as candidate identity.

D.3 Exact-amplitude physical realization

For a complete amplitude $A \in \mathcal{A}_{76}^{\text{phys}}$, define

$$D_A = (I - A^*A)^{1/2}, \quad D_{A^*} = (I - AA^*)^{1/2},$$

$$\mathbb{U}_{76}(A) = \begin{pmatrix} A & D_{A^*} \\ D_A & -A^* \end{pmatrix}.$$

The defect identity $AD_A = D_{A^*}A$ gives

$$\mathbb{U}_{76}(A)^*\mathbb{U}_{76}(A) = I, \quad \mathbb{U}_{76}(A)\mathbb{U}_{76}(A)^* = I,$$

and the observed transition block is exactly

$$P_f\mathbb{U}_{76}(A)P_i = A.$$

Thus

$$\text{RealAmp}_{76}(A) = (A, \mathbb{U}_{76}(A), \text{Own}_{76})$$

is an exact target-scale transition process. The defect blocks are complementary no-transition channels required by unitarity; the admitted D2 transition projection is the literal block A , and the exact base reduct removes the auxiliary completion data. The theorem does not identify this colligation with a unique ultraviolet completion. A raw compiled point is physically realized exactly where $A_{H,C}^{\text{tot}} \in \mathcal{A}_{76}^{\text{phys}}$, but the full enriched realization is not the colligation alone.

D.4 Constraint-preserving source-faithful physical amalgamation

For a semantic B15 physical representative W_H , the split-anchored D2 cell diagram $\mathbf{D}_{76}(H, C)$ contains the correction and hyperprocess decorations, the amplitude-sum relation, exact colligation, owner/matching/continuation maps, and every induced or higher D2 incidence. Its boundary is attached by closed split isometries to the complete retained anchor of W_H ; the base cells are identified, not copied. The full physical realization is the relative cell attachment

$$W_{H,C} = W_H \bigcup_{\mathbf{B}_{76}(W_H)}^{\text{phys}} \mathbf{D}_{76}(H, C).$$

The semantic residual packet is partitioned into inherited, pure-process, cross, and common-cone bands. The inherited band is the pullback of the already vanishing B15 residuals. The pure-process band is (8). The matching, continuation, vacuum/deformation, owner, causal, induced-effect, and higher-incidence

cross equations are exactly (9)–(16). Under the split direct-sum decomposition each residual has base, process, and mixed blocks; all three vanish. Therefore

$$W_{H,C} \in \text{PhysMod}_{4,D2}^{\text{sem}}, \quad \Pi_{76}^{\text{tr}}(W_{H,C}) = A_H + C, \quad \text{Res}_{\text{osc}}(W_{H,C}) \simeq W_H.$$

Its carrier trace is pointwise activity reduced:

$$\chi_{\text{car}}(W_{H,C}) \simeq P_{H,C}^{\text{red}} \simeq \begin{cases} H, & C = 0, \\ P_{H,C}, & C \neq 0. \end{cases}$$

Physical correction deletion is the split retraction,

$$\text{Del}_{c76}^{\text{phys}}(W_{H,C}) \simeq W_H, \quad \chi_{\text{car}} \circ \text{Del}_{c76}^{\text{phys}} \simeq U_{D2}^{\text{CMP}} \circ \chi_{\text{car}}.$$

Hence the formal reduct is the trace shadow of a physical restriction, and the common apex is a cone inside the semantic relation locus rather than merely a diagram carrying similarly named maps.

D.5 Complete fibred response

For one complete response frame η ,

$$K_{H,C,\eta} = \mathcal{R}_{76,\eta}(A_H + C), \quad \Gamma_{H,C,\eta} = \mathfrak{G}_{76,\eta}(K_{H,C,\eta}, K_{H,C,\eta}).$$

Lawful response arrows transport the exact amplitude response, Hermitian form, instrument, likelihood, and acceptance together. The marked affine family is selected only after the complete raw source domain and response groupoid exist:

$$C_{H,z}^* = -A_H + zA_\star, \quad A_{H,C_{H,z}^*}^{\text{tot}} = zA_\star.$$

No response-visible label appears in the source signature.

D.6 Exact nonclaims

The formal source extension does not assert that every microscopic theory contains a correction process, that every typed correction is independently physical, that the exact colligation is a unique ontology, or that AASC creates a missing source occupant. It declares one conservative target-complete comparison component whose raw candidates are evaluated by the same physical, response, standing, and failure rules as every intrinsic source chart.

E Family atlas

The following table summarizes the exact symbolic source-to-amplitude maps and survivor conditions. Each row abbreviates the full residual system in [section 9](#); it is not a finite registry substitute.

Band	Exact amplitude map	Exact solved object and complement
CMP	$A = A_H + C$, $C \in \mathcal{V}_{76}^{\text{src}}$; constraint-preserving $W_{H,C} \in \text{PhysMod}_{4,D2}^{\text{sem}}$ only where $A \in \mathcal{A}_{76}^{\text{phys}}$, with trace $P_{H,C}^{\text{red}}$ and D2 process projection $\text{RealAmp}_{76}(A)$.	Guarded locus of B15 survival, classwise vanishing inherited/pure/cross residuals, semantic common cone, physical base restriction, activity reduction, one complete response frame, defined kernel, accepted status, exact reduct; complement includes any nonzero amalgamation residual, non-contractive amplitude, frame failure, undefined/ROOS response, and acceptance failure.
LAW	$A = \bigoplus_{\lambda} \sum_{\pi} A_{\pi}^f L_{\pi} A_{\pi}^i$.	Guarded zero locus of B15 plus L1–L5, amplitude physicality, memory/order/matching, response, reduct, cone and status; quotient by the LAW groupoid.
LOCAL	$A = \bigoplus_{\lambda} \sum_{\pi} A_{\pi}^f V_r G \cdots G V_1 A_{\pi}^i + A^{\text{contact/ind}}$, with full $G = (\not{p} - M - \Sigma^{\text{ind}} + i0)^{-1}$.	Guarded zero locus of B15 plus N1–N5, representation/anomaly, vacuum, threshold, induced-term, matching, amplitude physicality and response; quotient by the LOCAL groupoid.
HID	$A = \bigoplus_{\lambda,r} A_{\lambda}^f P_{\text{vis}} \langle r V \Omega \rangle P_{\text{in}} A_{\lambda}^i$.	Guarded zero locus of B15 plus H1–H5, minimal hidden support, dilation/projection, memory/population, amplitude physicality, response, reduct and status; quotient by lawful dilation arrows.
COMP	$A = \bigoplus_{\lambda} \sum_{\pi} A_{\pi}^f D_{\pi}^f B_{\pi} D_{\pi}^i A_{\pi}^i$.	Guarded zero locus of B15 plus C1–C5, constituent retention, binding/derivation, support, matching, amplitude physicality, response and failure; quotient by constituent/source arrows.
SOL	$A = \bigoplus_{\lambda} \sum_{\pi} A_{\pi}^f \int \langle \phi_f \mathcal{O}_{76,\pi} \phi_i \rangle d\mu A_{\pi}^i$.	Guarded zero locus of B15 plus SO1–SO5, profile equation, finite support/energy, topology, stability/endability, amplitude physicality, response and continuation; quotient by profile arrows.
HYP	$A = \bigoplus_{\lambda} \sum_{\pi} \text{Contr}_{\pi}(\{A_i\}, \{T_e\}, \text{Glue})$.	Guarded zero locus of B15 plus Y1–Y7, all active constituent/higher relations, gluing, amplitude physicality, response, exact reduct, cone and status; quotient by decorated-hypergraph arrows.

F Overlap and codescent

For solved chart objects x_i , the direct finite overlap residual is

$$\mathcal{E}_{i_0 \dots i_r}^{\text{high}} = (\Delta p^{\text{base}}, \Delta[A]_{\text{St}}, \Delta \overline{\text{Resp}}_{76}, \Delta K_\eta, \Delta \Gamma_\eta, \Delta \text{Stat}, \\ \Delta \text{Att}, \Delta \text{Match}, \Delta \text{Cont}, \Delta \text{Comp}, \Delta \partial \text{Stat}, \\ \Delta \text{High}, \text{Cocycle}, \text{Cone}).$$

Every response comparison occurs in one common lawful complete frame after transport. Every amplitude comparison is made in the fibred standing-amplitude quotient, not by literal raw-amplitude equality. The overlap is the zero locus of this residual modulo the diagonal lawful action. Its first nonzero coordinate is the exact separator or emptiness certificate.

All 28 unordered chart-band pair types with repetition are included by the same complete-core equalizer. The CMP rows use the general raw amplitude $A_H + C$; the marked family $zA_\star$ is only a solved subchart. The finite higher residual retains primitive higher incidence and cocycle data directly, so pairwise nonemptiness is never promoted into higher coherence. Codescent is formed only after source coverage, chart soundness, every pair equalizer, and direct higher coherence are established.

G Theorem dependencies and outputs

ID	Imported support	New local proof	Downstream export
D2-01	Target specification; experimental and theoretical sources	Complete mechanism-neutral response, applicability and status grammar	Fixed target object and update boundary
D2-02	PCR-3; PCR-5B B15; fixed source specification	Intrinsic plus complete C -indexed compiled source universe, finite hybrids, no-backflow mutation, and conservative reduct	Complete raw source charts and frames
D2-03	PCR-4 one-point discipline; SOE-D2-1	Constraint-Preserving Physical Amalgamation Theorem, activity-reduced trace, Retained-Oscillation Transparency and Base-Conservativity, downstream affine witness, exact transition block, semantic common cone, deletion naturality, response inheritance and active continuum interval	Nonempty continuum subfibre over every PCR-5B/B15 physical base realization
D2-04	PCR-5C witness retention/no merger; SCIC	In-situ retained standing restriction, transparency and frame compatibility	ρ_{D2} , no merger
D2-05	PCR-5C homotopy fibres	Seven exact chart objects, exact complements and interval injection	Exact fibre types and cardinal lower bounds
D2-06	SCIC; response covariance	Parameter-role exhaustion and NME nonpromotion	Descended physical response classes
D2-07	PCR-5C S/R/P/Y calculus	Exact chartwise identity and target-survival deletion strata	Exact S/R/P/Y atlas

ID	Imported support	New local proof	Downstream export
D2-08	D2-03/D2-04	Transparent marked survivor over arbitrary B15 frame	Full base image
D2-09	SCIC relative separators; refinement addendum	Universal split, exact separator export and coordinate diagnosis	Conjugation/LNV ledger entry
D2-10	PCR-5B/PCR-5C atlas methods	Twenty-eight pair systems plus direct finite higher residual and codescent	Exact global D2 groupoid
D2-11	PCR-4 applicability discipline	Total D2 status partition and re-opening map	Certified complements
D2-12	PCR-5C joint-target interface	Complete target/source/fibre/-composition/overlap export datum	PCR-5E marginal input

H Reproducibility and evidentiary boundary

Supplementary materials include:

- the LaTeX source and publication PDF;
- the exact public LEGEND CSV files;
- scripts reproducing the figures and integrity checks;
- SHA-256 manifests and derived-data checks;
- the fixed target/source specification and the theorem plan underlying the analysis;
- machine-readable summaries of the endpoint, standing coordinates, fibre dispositions, and theorem status.

The supplementary scripts reproduce file-identity checks, row counts, derived exposure, figure inputs, and endpoint records. They do not establish source completeness, exact-amplitude physicality, constraint-preserving split-anchored attachment, activity reduction, physical base restriction, one-point survival, SCIC separation, fibre exhaustion, or no merger. Those claims rest on the semantic argument in the manuscript.

I Exact scope and nonclaims

The following statements are explicitly outside the theorem:

1. an observed $0\nu\beta\beta$ event or nonzero realized LNV amplitude;
2. exact lepton-number conservation from a null search;
3. universal Dirac or Majorana neutrino character;
4. universal reduction of D2 to $m_{\beta\beta}$;
5. uniqueness of an NME, EFT chart, UV completion, or source architecture;
6. exact cardinality of $Q(\hat{T}_{D2}^{\text{Ge}})$ or any upper cardinal independent of the comparison universe;
7. global product factorization of D2 and oscillation or a common-generator theorem;
8. any D3 heavy-state/nonunitarity, D4 cosmological, D5 symmetry, or PCR-5E joint-target result;

9. multi-isotope closure or replacement of the ^{76}Ge target by a global $0\nu\beta\beta$ likelihood;
10. denial of local-QFT black-box/radiative consequences or identification of the universal target-scale process chart with a unique local UV completion;
11. a universal inference that deleting one D2 source block yields zero response or preserves target survival;
12. promotion of manifests, registries, hashes, or finite tests into semantic proof.

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